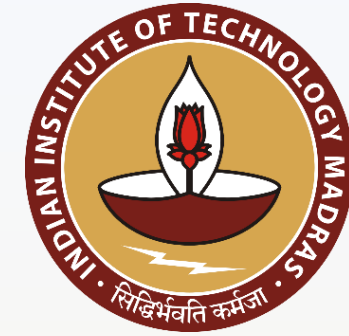


Seminar on



Towards an early detection of atherosclerosis in the human coronary artery using ultrasound contrast agent.

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(ME21S040)**

**Guided By
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Dr. S. Vengadesan (AMBE)**

Outline

- Introduction.
- Literature Survey.
- Literature Gap.
- Physical and Computational Domain.
- Numerical Methodology.
- Numerical Results and Discussion.
- Experiments Setup and Sample preparation.
- Experiments Results and Discussion.
- Conclusion.
- Reference.

Introduction: Coronary Artery

- The Coronary Artery supplies blood to heart muscles. It is divided into two parts: Left and Right coronary artery.
- The average length of the left anterior descending artery is **100-130 mm (Waller et al. 1992)**.
- The average Lumen diameter is **4.6 mm (Douglas et al. 1988)**, and the average thickness of the wall is around **1 mm (Perry et al. 2013)**.
- Time average velocity in left anterior descending artery **6.8 ± 4.3 cm/s** and in Right coronary artery is **6.0 ± 1.6 cm/s (Johnson et al. 2008)**.
- Pressure fluctuation in coronary arteries for different age groups are: **(Pitt et al. 1969):**

Human Age (year)	40	48
Pressure (mm of Hg)	90-96	93-96

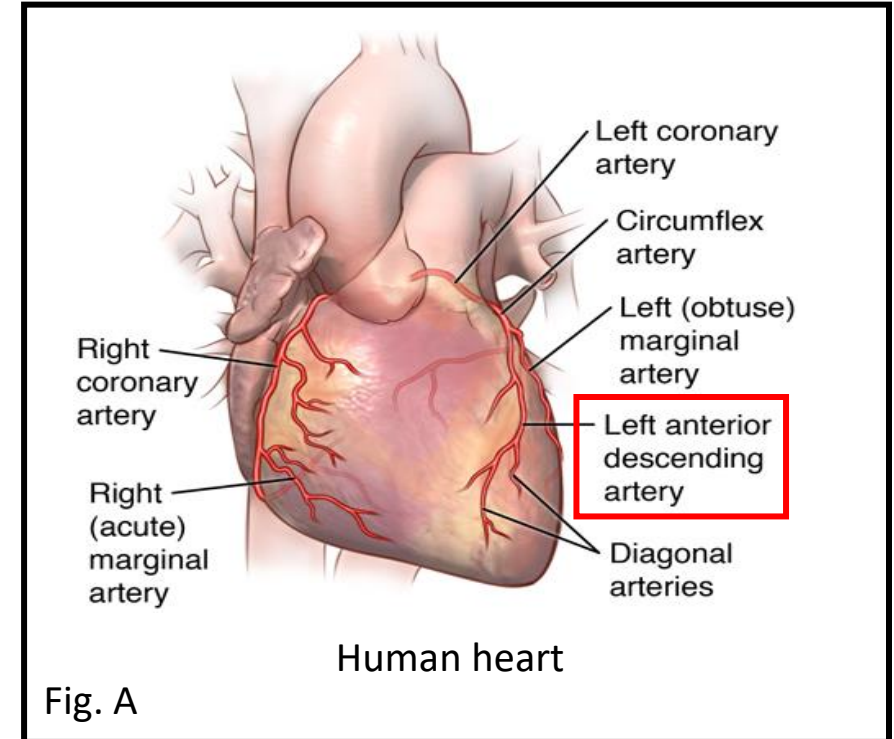


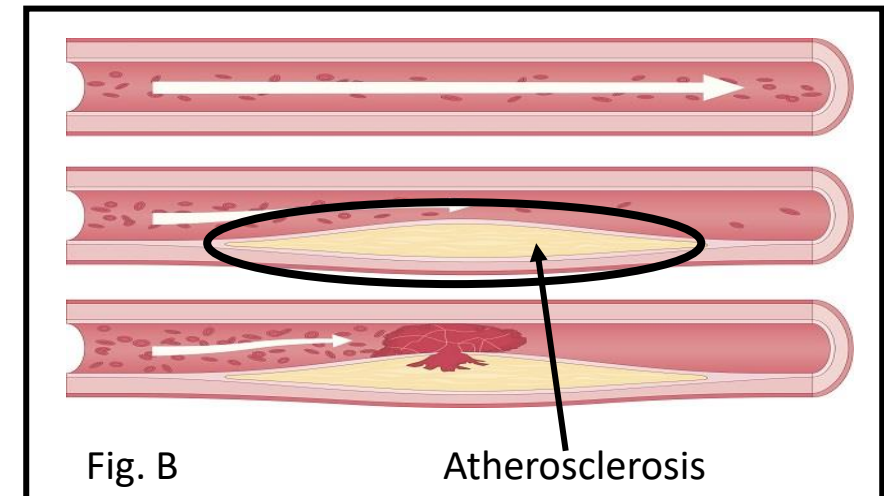
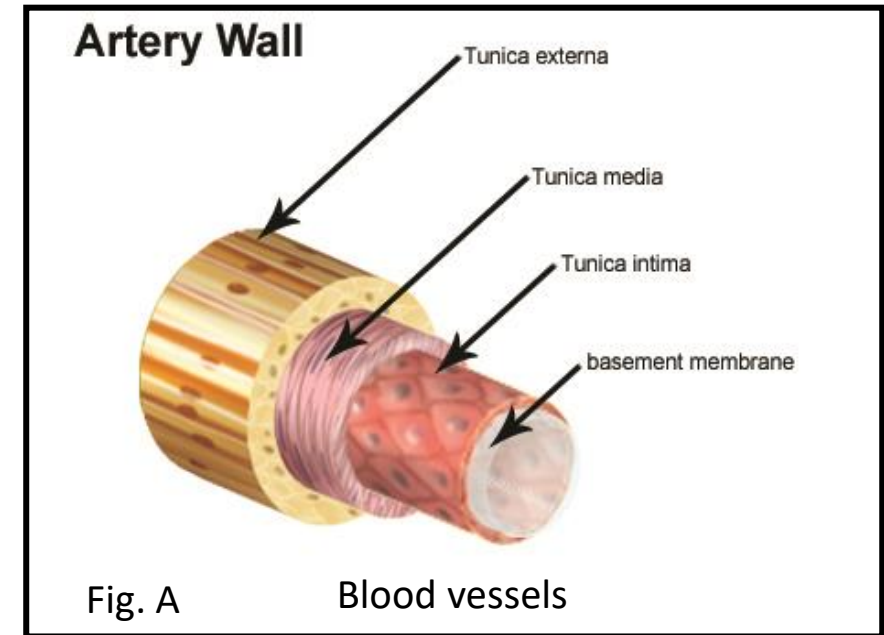
Fig. A Ref: <https://www.hopkinsmedicine.org/health/conditions-and-diseases/anatomy-and-function-of-the-coronary-arteries>

Introduction: Atherosclerosis

- It is a build-up of fat between tunica intima and tunica media.
- It is categorised into two parts: stable and unstable (vulnerable) plaque.
- The main component is lipid (kind of fat) accumulation.
- Vulnerable plaque has a thin fibrous cap (cap thickness is 65 to 150 micrometres) and a large lipid core, which may break from the plaque and block the blood passage.
- The average length of atherosclerosis is **$12.16 \pm 5.61\text{mm}$** (Yu et al. 2017).
- The mean cross-section area of atherosclerosis is **5.55 mm^2** (Ge et al. 1994).

1. Fig. A Ref: <https://training.seer.cancer.gov/anatomy/cardiovascular/blood/classification.html>

2. Fig. B <https://www.heartandstroke.ca/heart-disease/conditions/atherosclerosis>

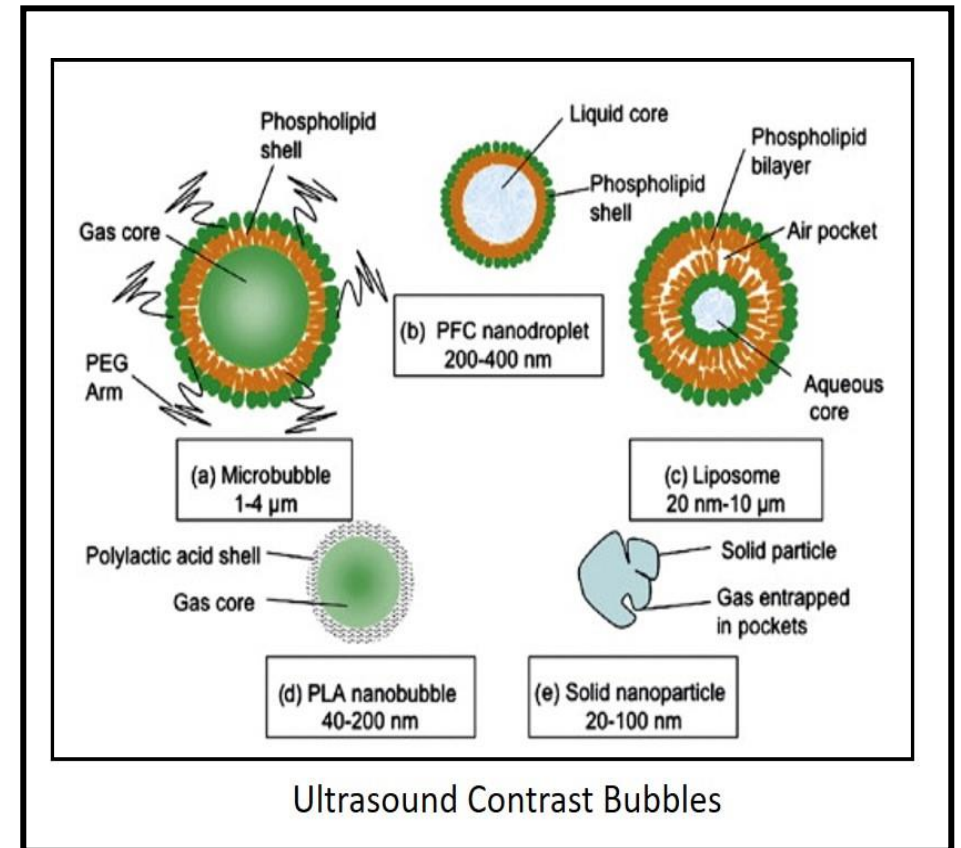


Literature Survey: Vulnerable Atherosclerosis

S. NO	Description	Reference
1	Vulnerable plaque has a large lipid core and thin fibrous cap (65 to 150 μm).	Fayad and Fuster (2001)
2	Axisymmetric and non-axisymmetric stenosis show very different wall shear stress.	Stroud et al. (2000)
3	Skin friction increases significantly with increased stenosis length for a given yield stress value	Venkatesan et al. (2013)
4	Methods available for the diagnosis of atherosclerosis, such as; ECG, MRCA, ultrasound, CT SPECT, and IVUS.	Poznyak et al. (2023)
5	In many cases, it is not detected by X-ray angiography.	
6	The cost of available diagnosis techniques like angiography and intravascular ultrasound imaging (IVUS) is very high.	Zhou et al., (2021).
7	The penetration depth of ultrasound pulse decreases with increasing frequency; the artery located at a greater depth in the body, like the coronary artery, can not be imaged early using these high frequencies.	de Korte et al. (2011)
8	The acoustic impedance of soft tissues (like blood vessels, fat, etc), blood, and air are 1.63 MRayls, 1.61 MRayls, and 0.0004 MRayls respectively.	Tole (2005)

Literature Survey: Contrast Bubbles

- Contrast agent is also known as contrast media, which is used to increase the acoustic impedance and radiodensity inside the body.
- Bubbles are categorized into two parts: Encapsulated and non-encapsulated. The main function of encapsulation is to develop steric hindrance, prevent gas from flowing out, and decrease the dissolving rate in the blood (**MacDonald et al. 2004**).
- The diameter of the bubble is around **1 to 5 microns** (**Chong et al. 2018**), and the blood vessel capillary diameter is around **8 to 10 micrometres**.
- Contrast agents like Definity (Lantheus Medical Imaging), Optison (GE Healthcare), Sonazoid (GE Healthcare), and SonoVue/Lumason (Bracco) are available commercially.



Ref: <https://images.app.goo.gl/Bgz1bECmL7Cb8oFTA>

Literature Survey: US Imaging

- Echo to convert reflected sound energy into image, B-mode imaging.
- Reflection and refraction, scattering, absorbing, attenuation.
- The strength of echo will be high if the difference in acoustic impedance is high at the interface.

$$R = \left(\frac{Z_2 - Z_1}{Z_2 + Z_1} \right)^2 \quad T = \left(\frac{2 * Z_2}{Z_2 + Z_1} \right)^2$$

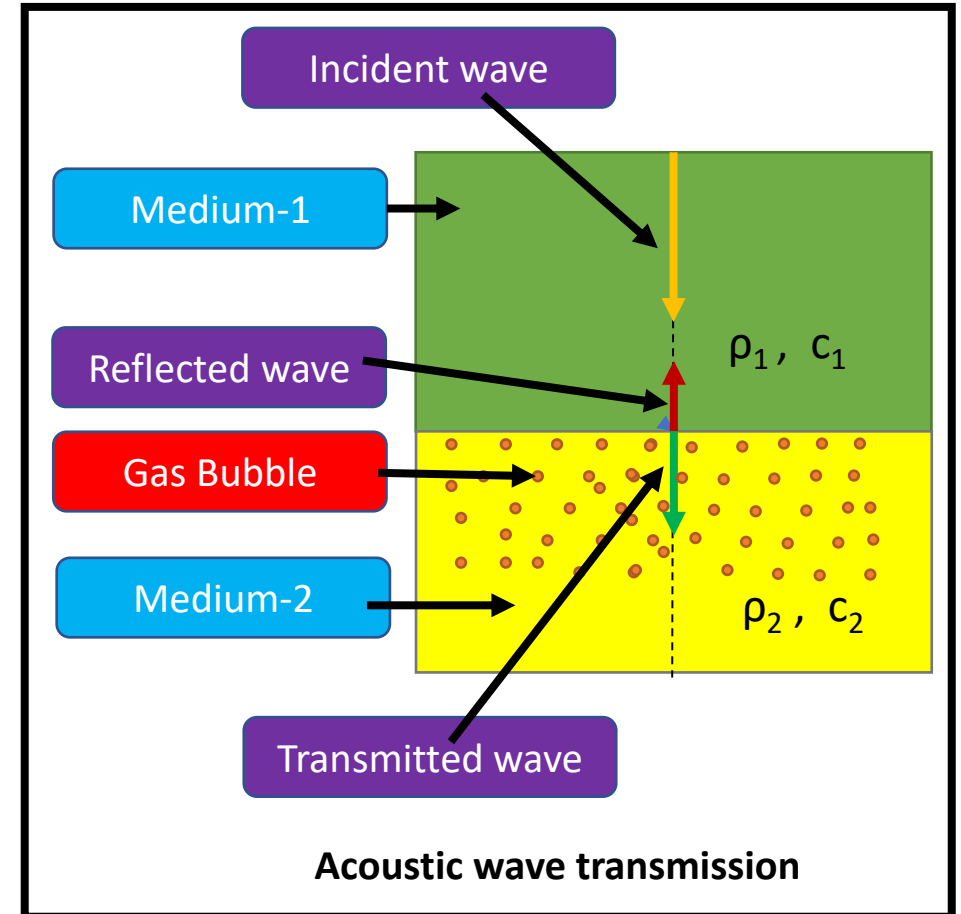
- $MI = \frac{P}{\sqrt{F}}$, (Apfel et al. 1991). MI should be less than 1.9, but most imaging is being done around MI is 1 (Averkiou et al. 2020).

P = Amplitude Pressure (MPa).

F = Frequency (MHz).

R= reflected Intensity ratio, T = transmitted intensity ratio, $Z = \rho c$, Z= acoustic Impedance

ρ = density (kg/m³), c = sound speed (m/s).



Literature Survey: Contrast US

S. NO	Description	Reference
1	Ultrasound imaging always compromises between resolution and penetration; resolution increases with increasing frequency, but the depth of penetration decreases. A resolution of 500 μm is obtained with 5 MHz, which is improved to 125 μm for a 20 MHz pulse, but the depth of penetration decreases.	de Korte et al. (2011)
2	Contrast agents are currently clinically approved to image cardiac structure; with advancement, it is also being used for imaging of the liver, carotid, breast, and abdomen.	Helfield (2019)
3	Conventional ultrasound method can detect the velocity but it does not tell in which direction blood is flowing but using contrast agent, ultrasound can predicts the direction of blood flow.	Frinking et al. (2000)
4	It has a better advantage in producing real-time non-cardiac imaging than CT and MR imaging where the mechanical index is less than 0.4.	Chong et al. (2018).
5	It enhanced the visualization of the c-IMT (carotid intima-media thickness), and the result is consistent with the MRI result.	Feinstein (2006)
6	It is used for the detection of intracardiac thrombus and the visualization of wall motion, and it is also used for imaging of the carotid and aorta.	Chinkel et al. (2016)

Gap in Literature

- Major work has not been done for unsymmetric growth of atherosclerosis [or growth of plaque from one side of blood vessels] in 3 dimensions, but in a natural case, growth of atherosclerosis is highly unsymmetric and from one side of blood vessels.
- Modelling of atherosclerosis in coronary arteries has not been done for in-vitro experiments.
- Work has not been done for early detection of atherosclerosis in the coronary arteries using ultrasound contrast agents.

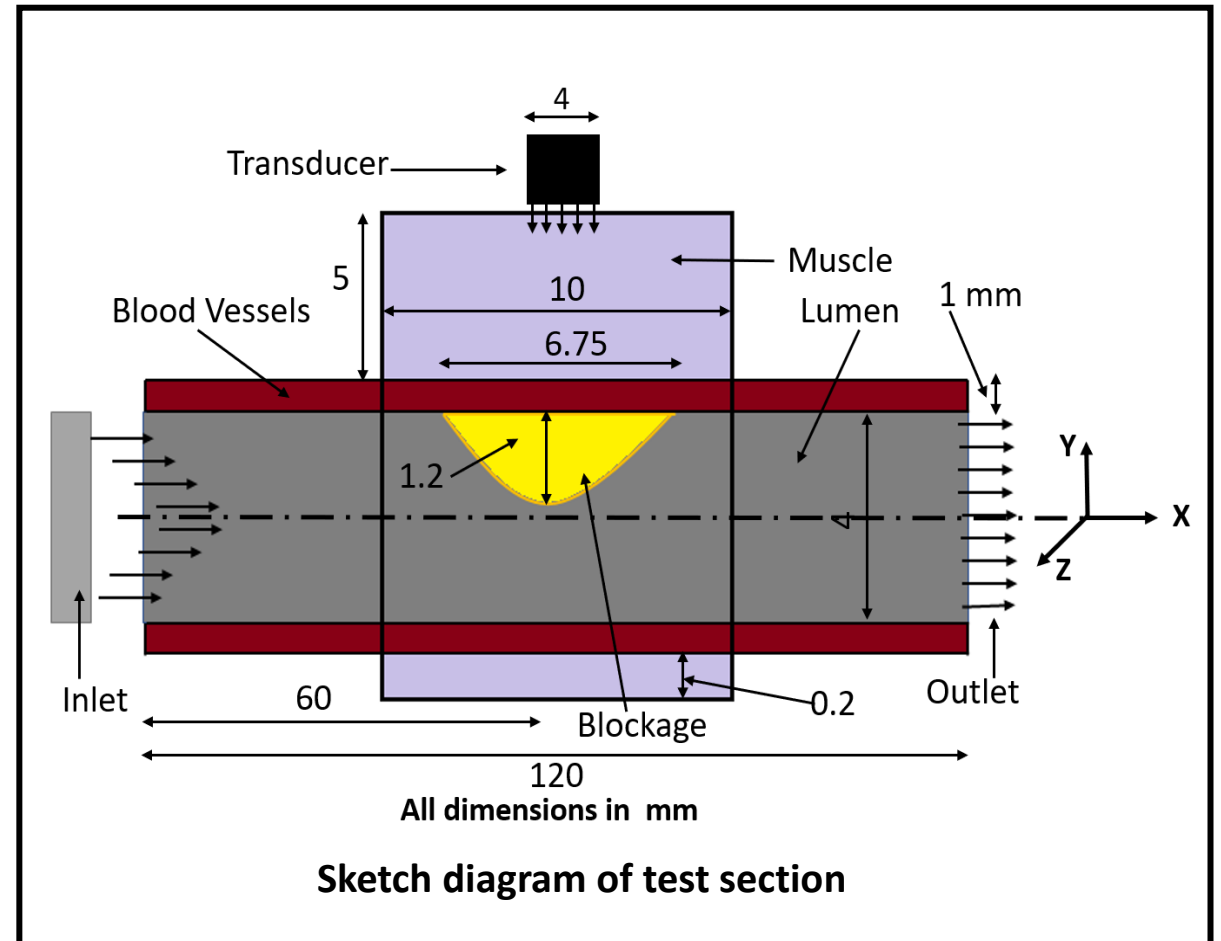
Objectives and scope

Main objective: To develop a methodology which can improve the ultrasound imaging technique using ultrasound contrast microbubbles for the detection of early-stage atherosclerosis.

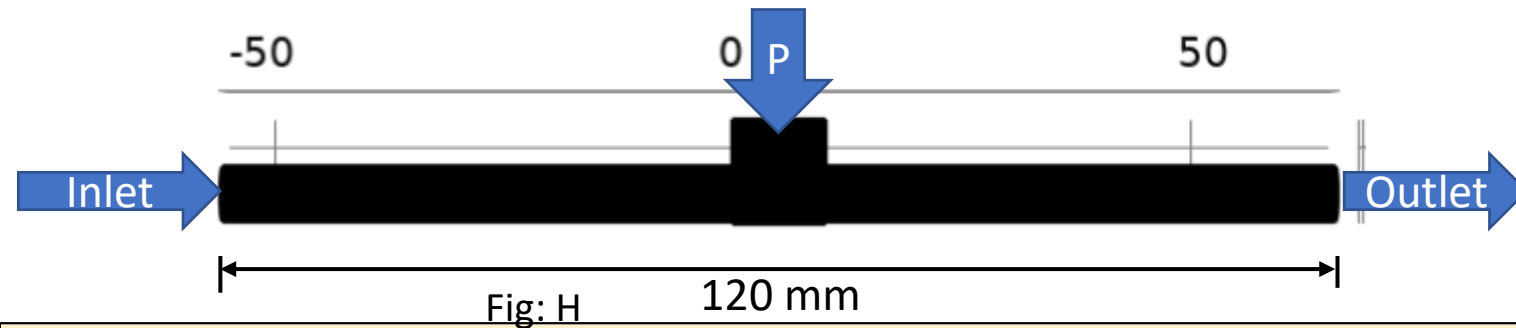
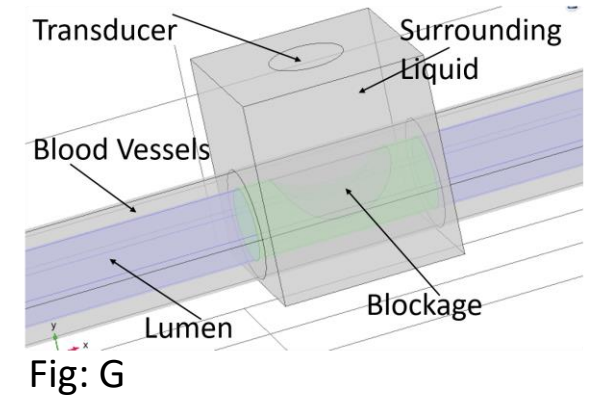
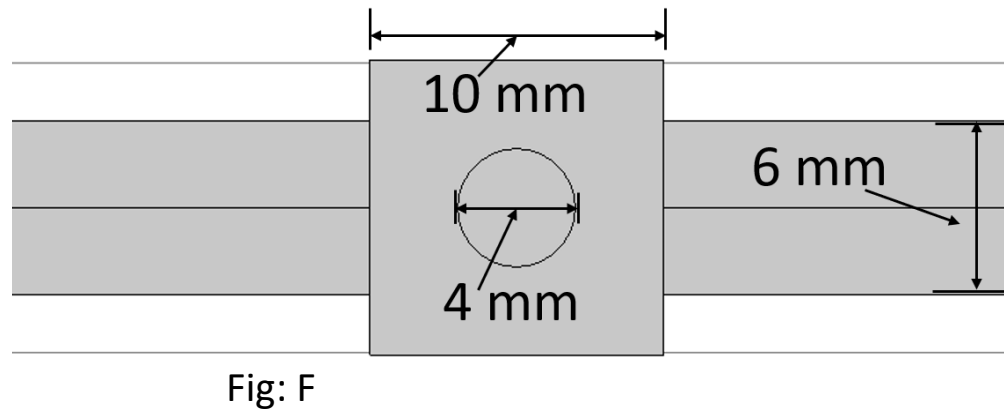
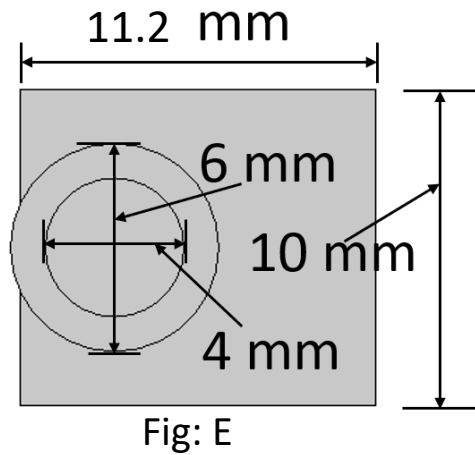
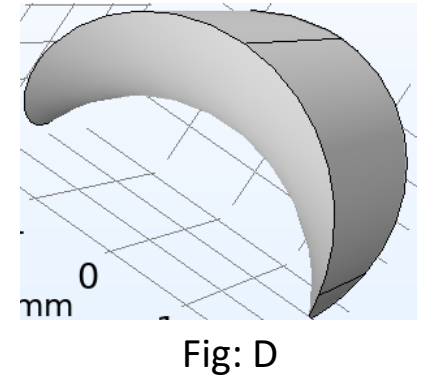
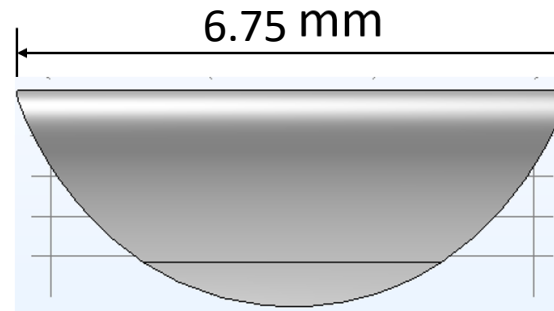
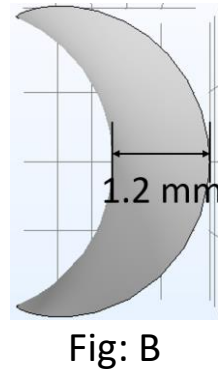
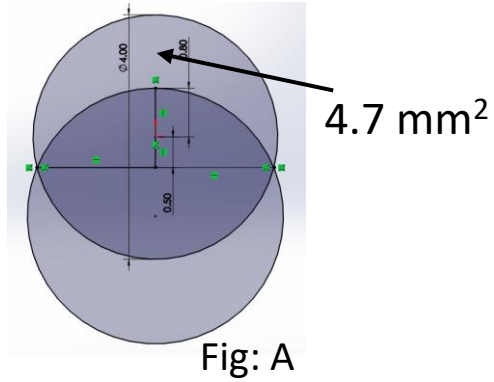
- We are doing a numerical simulation in which we are considering the dimensions and parameters of the coronary artery (LAD) and atherosclerosis to design the physical domain.
- We are doing the in-vitro experiments in which we are considering the dimensions and parameters of the coronary artery (LAD) and atherosclerosis to design the experiments set up and test section with atherosclerosis.
- We are also producing free air gas bubbles as contrast agents, whose diameter is 2 μm .

Physical and Computational Domain

- The circular diameter of the transducer is 4.
- Length of atherosclerosis is **6.75 mm (Yu et al. 2017)**.
- Length of blood vessels is **120 mm (Waller et al. 1992)**.
- The thickness of the blood vessel wall is **1 mm (Perry et al. 2013)**.
- Maximum depth of atherosclerosis is 1.2 mm, and maximum cross-section area is **4.7 mm² (Ge et al. 1994)**.
- Lumen diameter is **4 mm (Douglas et al. 1988)**.
- Flow is from left to right.
- Muscles dimensions are width= 10 mm, length= 10 mm, and height= 11 mm.



Physical and Computational Domain



The minimum length of grid size required to resolve acoustic wave is $\lambda/5$ for 2nd-order Lagrange elements.

Numerical Methodology - Equation

Continuity Equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho u) = 0$$

Momentum Equation, Liquid Phase

$$\rho_l \frac{\partial u_l}{\partial t} + \rho_l (\nabla \cdot u_l) u_l = \nabla \cdot (pI + K) + \phi_l \rho_l g + F$$
$$K = \mu_l (\nabla u_l + (\nabla u_l)^T)$$

Transport Equation

$$\frac{\partial (\rho_g \phi_g)}{\partial t} + \nabla \cdot (\rho_g \phi_g u_g) = 0$$

ρ = Mixture Density, μ_B = Bulk Viscosity, u_l = liquid phase velocity, ρ_g = Gas Density, ρ_l = Liquid Density, ϕ_l = Liquid Fraction, ϕ_g = Gas Fraction, u_g = Gas phase velocity, p_t = Acoustic Pressure, q_d = acoustic dipole domain source

Pressure Acoustic Equation, Viscous Model

$$\frac{1}{\rho c^2} \frac{\partial^2 p_t}{\partial t^2} + \nabla \cdot \left(\frac{-1}{\rho} \right) (\nabla p_t - q_d) + \frac{\delta}{\rho c^2} \frac{\partial \nabla p_t}{\partial t} = Q_m$$
$$\delta = \frac{1}{\rho} \left(\frac{4\mu}{3} - \mu_B \right)$$

Pressure Acoustic Equation, Elastic Model

$$\frac{1}{\rho c^2} \frac{\partial^2 p_t}{\partial t^2} + \nabla \cdot \left(\frac{-1}{\rho} \right) (\nabla p_t - q_d) = Q_m$$

Mixture Model

$$\rho = \phi_l \rho_l + \phi_g \rho_g$$
$$\mu_B = \phi_l \mu_{Bl} + \phi_g \mu_{Bg}$$

Numerical Methodology - BC

Sound velocity in the homogeneous mixture of water and air (Wallis 2020).

$$\frac{1}{c^2} = (\alpha_1 \rho_1 + \alpha_2 \rho_2) \left(\frac{\alpha_1}{\rho_1 c_1^2} + \frac{\alpha_2}{\rho_2 c_2^2} \right)$$

c = sound mixture speed, α_1 = volume fraction of water, α_2 = volume fraction of air, ρ_1 = density of water, ρ_2 = density of air, c_1 = speed of sound in water, c_2 = speed of sound in air.

Table 4: Effect of volume fraction of air bubble on acoustic impedance [Z].

Air bubbles VF	Speed [m/s]	Density [kg/m ³]	Z [MRayl]
0.001	356.39	996.004	0.35
0.002	255.78	995.008	0.25
0.003	209.78	994.012	0.20

Acoustic impedance [Z] = speed X density.

Table 1: Boundary conditions are used for solving continuity and momentum equations.

Fluid	Inlet	Outlet
Water	$2 * u_{avg} (1 - r^2/R^2)$ m/s	100 mm of Hg
Air volume fraction	0.001	Gradient=0

$u_{avg} = 6 \text{ cm} = 0.06 \text{ m/s}$ (Johnson et al. 2008).

Table 2: Boundary conditions used in solid mechanics equation.

Side Surface	Outer Surface	Inner Surface
Fixed	0 mm of Hg	Fluid Pressure, FSI

FSI = Fluid-Structure Interaction

Table 3: Boundary conditions used in pressure acoustic equation.

Transducer imposed surface	Other surface
Acoustic Pressure= 1 MPa	Acoustic Impedance gradient= 0

Numerical Methodology - Materials

Table 1: Materials and their properties used to mimic the human organ parts and in numerical simulations

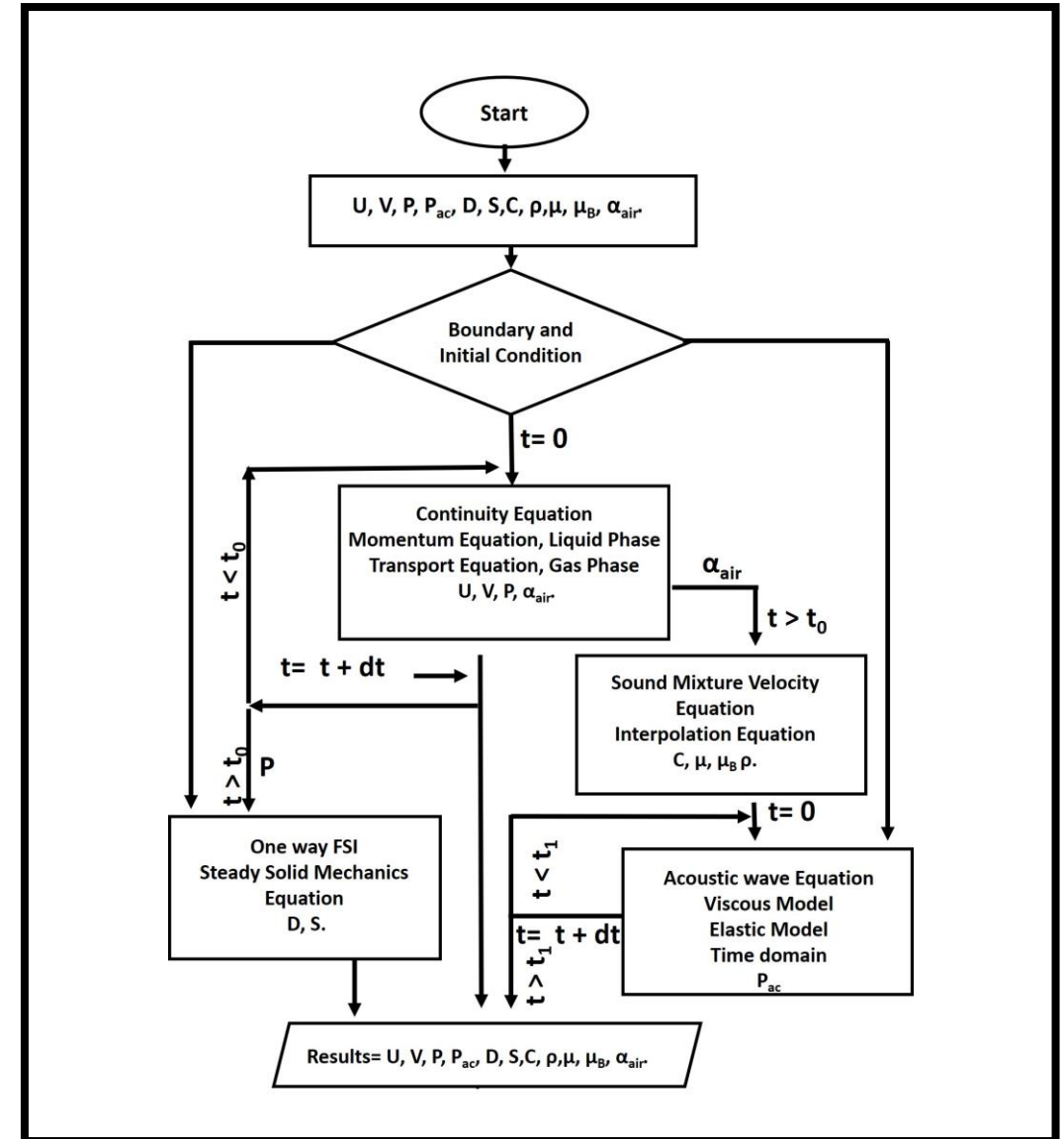
S. No	Name	Mimicking	Density (kg/m ³)	Young's Modulus	Poisson Ratio	Sound speed (m/s)	Acoustic Impedance (MRayls)
1	Silicone Tube	Blood vessels	1000	2.94 MPa	0.47	1000	1
2	Agar-Agar Powder	Atherosclerosis	1040	52 kPa	0.32	1500	1.56
3	Free air bubbles	Contrast agents	1.16	-----	-----	340	0.0004
4	Saline Water	Blood	1000	-----	-----	1496.73	1.5

- Bulk viscosity and dynamic viscosity of water, respectively, are 0.25×10^{-2} Pas, 0.89×10^{-3} Pa s (**Holmes et al., 2011**) and for air, these are 0.96×10^{-5} kg/(m s) and 2.27×10^{-5} kg/(m s), respectively (**Bruno et al., 2023**).
- Young's modulus and Poisson ratio for silicon are taken from **Muslov et al. 2021**. The properties of agar-agar powder are also taken from the literature. Its Young's modulus and Poisson ratio are taken from **Ross and Scanlon (1999)** and density and sound speed are taken from **Zell et al. (2007)**.

Numerical Methodology - Algorithm

- Declare the variable
- Impose the boundary condition and initialize with the initial condition.
- Solve the continuity equation, momentum equation, and transport equation.
- Impose the static blood pressure on the blood vessels and solve the solid mechanics equation for the steady case.
- Solve sound mixture velocity equation.
- Solve acoustic equation

Software: COMSOL 6.0



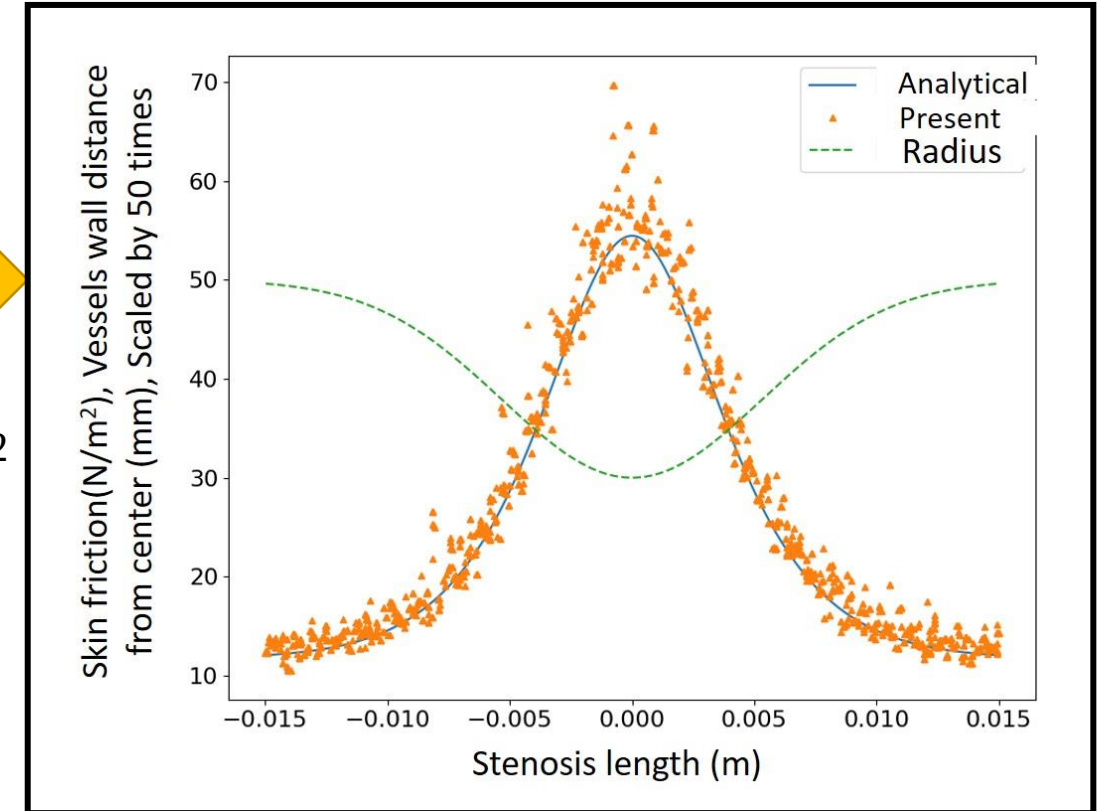
Numerical Methodology - Validation

[1]. Mathematical Analysis of Casson Fluid Model for Blood Rheology in Stenosed Narrow Arteries, Venkatesan et al. (2013)

Length of vessels	5cm
Radius	1mm
Stenosis Length	30mm
Stenosis Shape	$R(z) = R_0(1 - a * \epsilon^{(-bz^2)})$

a =stenosis Depth/ R_0 , L_0 = Stenosis Length, $b=m/L_0^2$, m = Shape Factor = 2

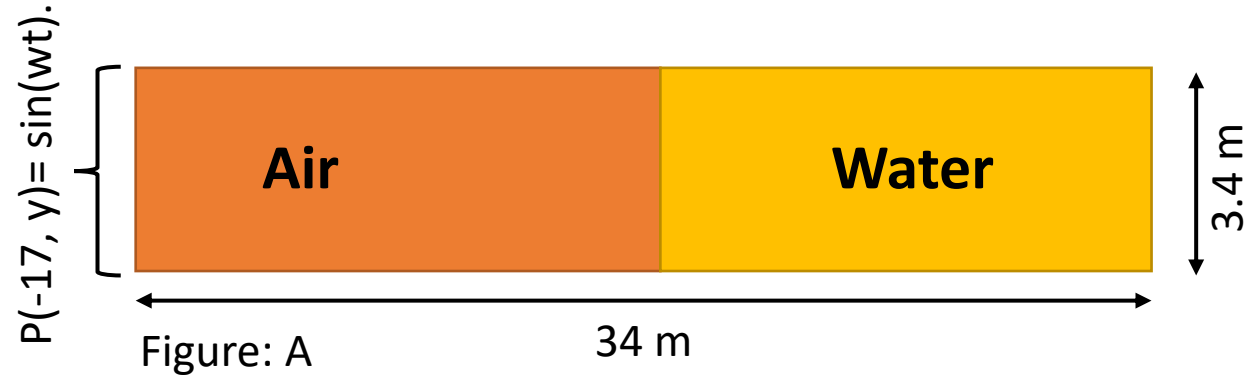
S. No	R_i	R_o	T	Simulation (MPa)	Wibawa et al. (2020) (MPa)
1	55	61	6	92.7	92.59
2	54	61	7	68	67.93
3	53	61	8	51.7	51.72
4	52	61	9	40.4	40.47
5	51	61	10	32.2	32.33



[2]. Stress Analysis of Thick-Walled Cylinder for Rocket Motor Case under Internal Pressure, Wibawa et al. (2020).

Density = 2700 kg/m³, Young's Modulus = 68.9 GPa, Internal Pressure = 2 MPa, Poisson's ratio = 0.33, Length = 300 mm

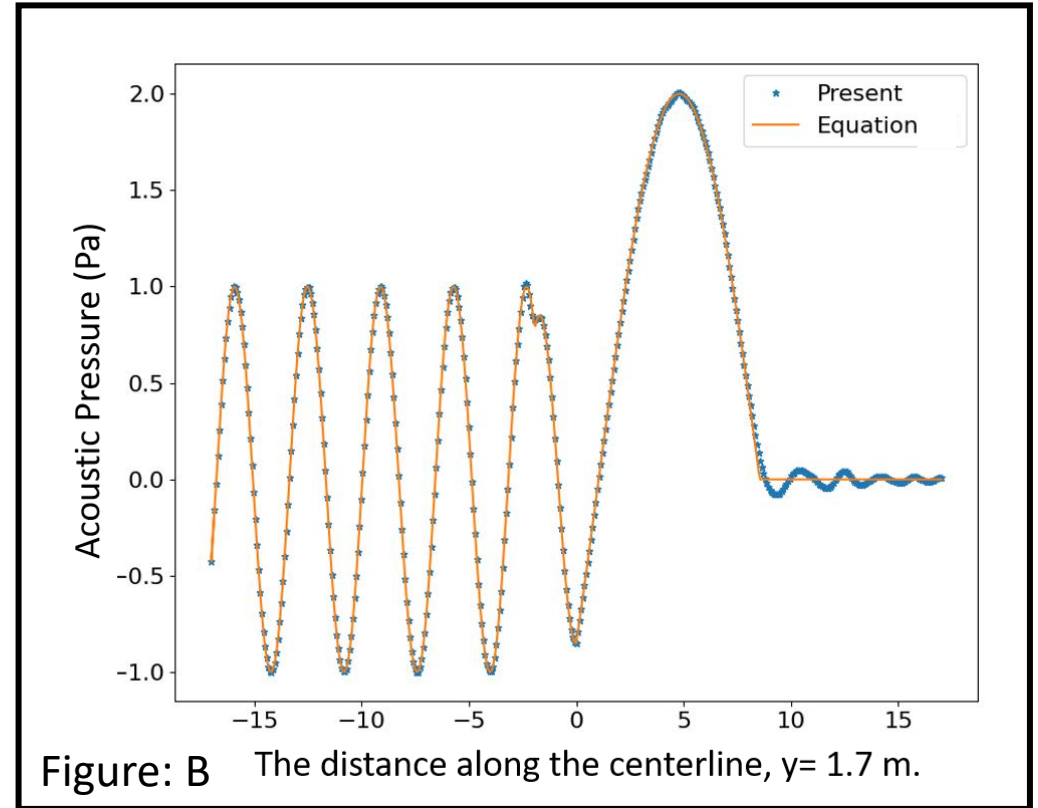
Numerical Methodology - Validation



$$p(0.05s < t < 0.06133s) = F(x)$$

$$F(x) = \begin{cases} \sin(wt - k_1x) & x < 17 - 340t \\ \sin(wt - k_1x) + \frac{\rho_2 c_2 - \rho_1 c_1}{\rho_2 c_2 + \rho_1 c_1} \sin(wt + k_1x) & 17 - 340t \leq x < 0 \\ \frac{2\rho_2 c_2}{\rho_2 c_2 + \rho_1 c_1} \sin(wt) & x = 0 \\ \frac{2\rho_2 c_2}{\rho_2 c_2 + \rho_1 c_1} \sin(wt - k_2x) & 0 < x \leq 1500t - 75 \\ 0 & 1500t - 75 < x \end{cases}$$

$k_1 = w/c_1, k_2 = w/c_2, w = \text{angular velocity}, x = \text{distance along } x\text{-axis}.$



[3]. You, X., Yin, J., Yao, Y. and Li, W., 2024. Time-domain acoustic wave propagations in multi-fluids using a weak-form meshfree method. *Ocean Engineering*, 292, p.116531.

Numerical Results-Grid-Coronary Artery Domain

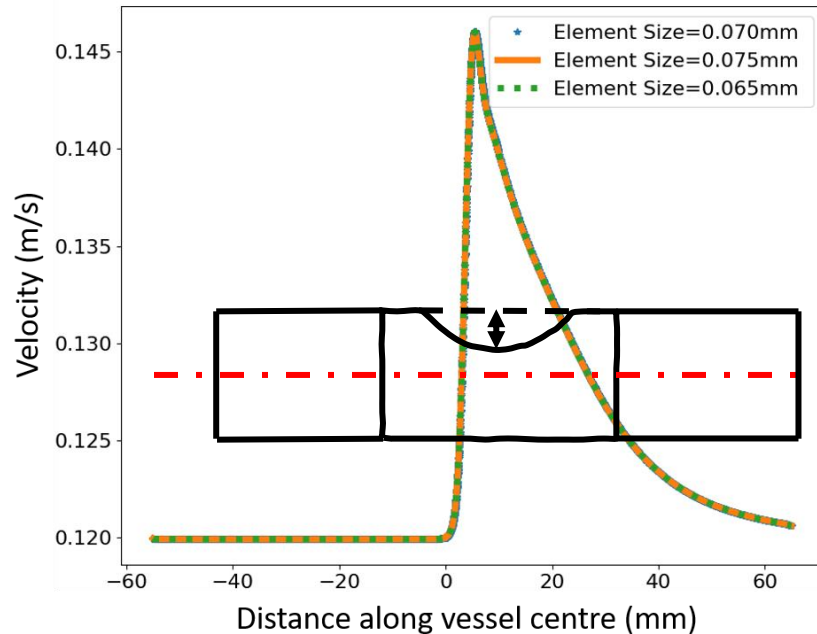


Fig A: Variation of velocity along the centerline of blood vessels for different grid sizes.

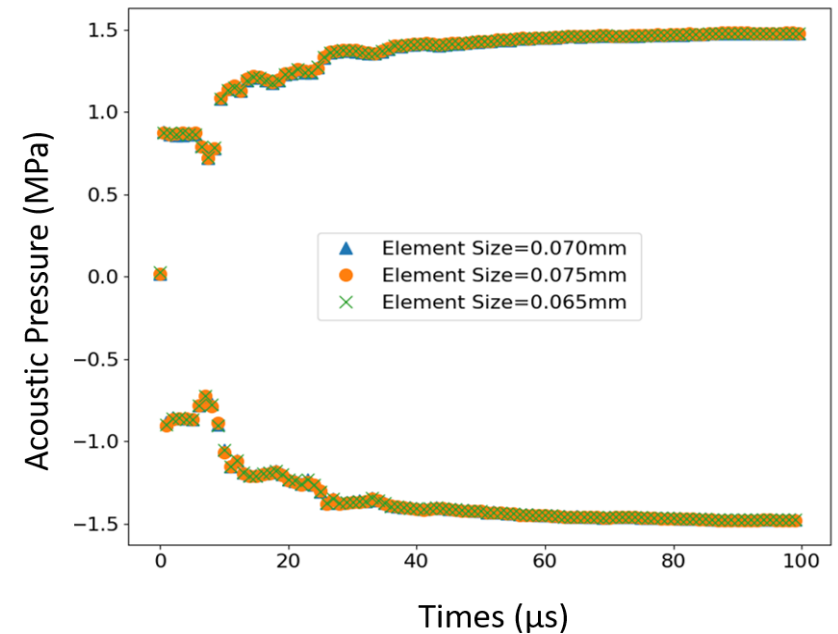
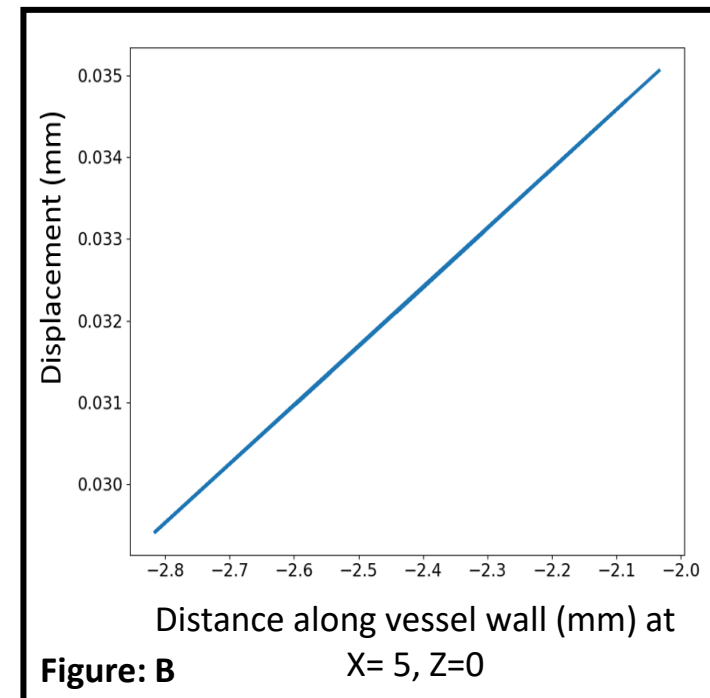
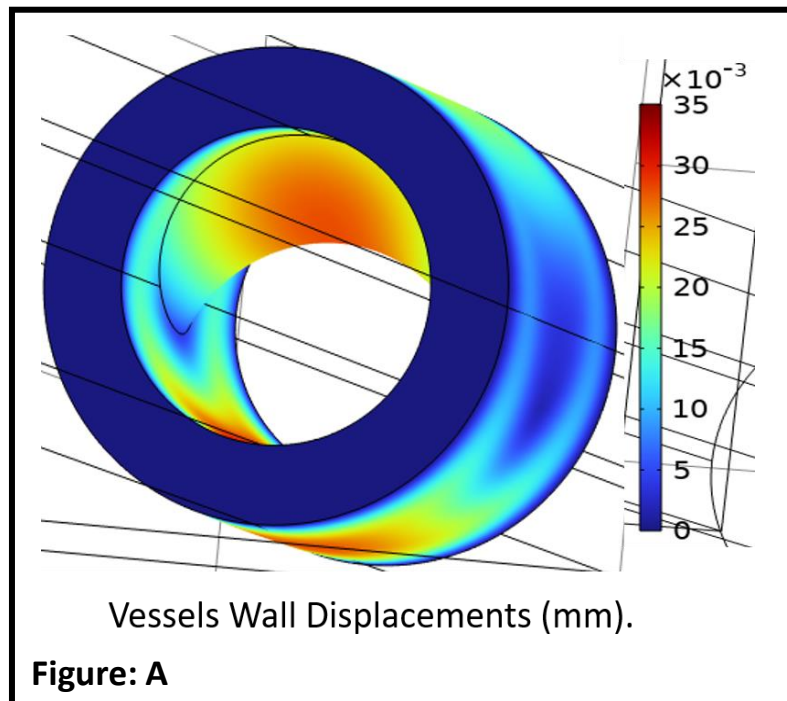


Fig B: Strength of echo monitor at point (5,7,0) in the presence of the contrast bubble.

- The minimum length of grid size required to resolve acoustic wave is $\lambda/5$ for 2nd-order Lagrange elements.

Numerical Results and Discussion-Vessels wall

- The contour plot shows that the distribution of the shear rate is not uniform.
- The shear rate is high on the blockage surface.
- The displacements vary linearly along the radial direction of the blood vessels.
- The maximum displacement of the inner wall of blood vessels is $35\text{ }\mu\text{m}$.



Numerical Results and Discussion- Flow

Contour: Velocity fields in the x-direction, U_x (m/s), time $t = 4s$.

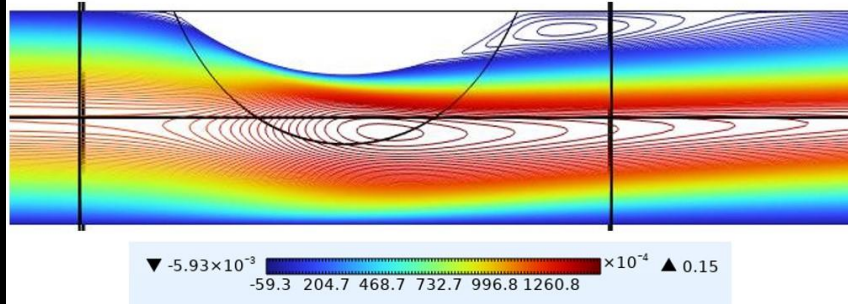


Fig A: Contour plot of velocity

Flow Direction

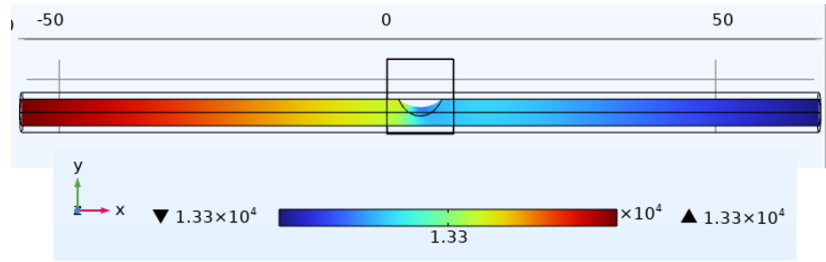
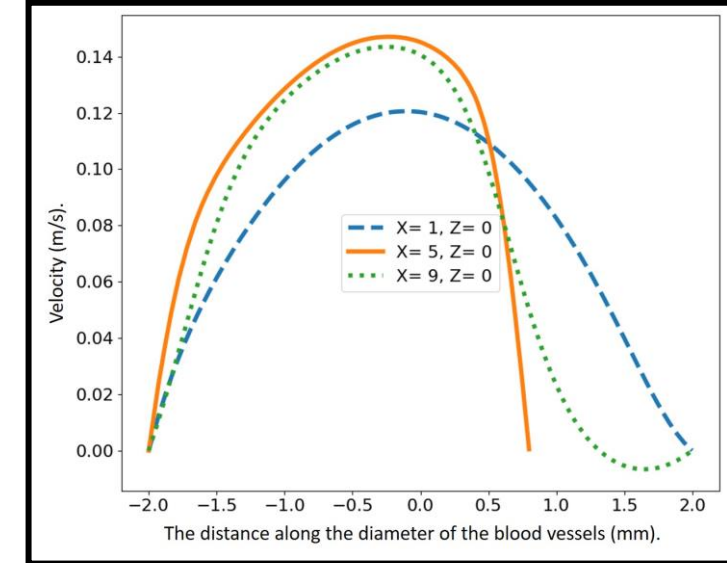
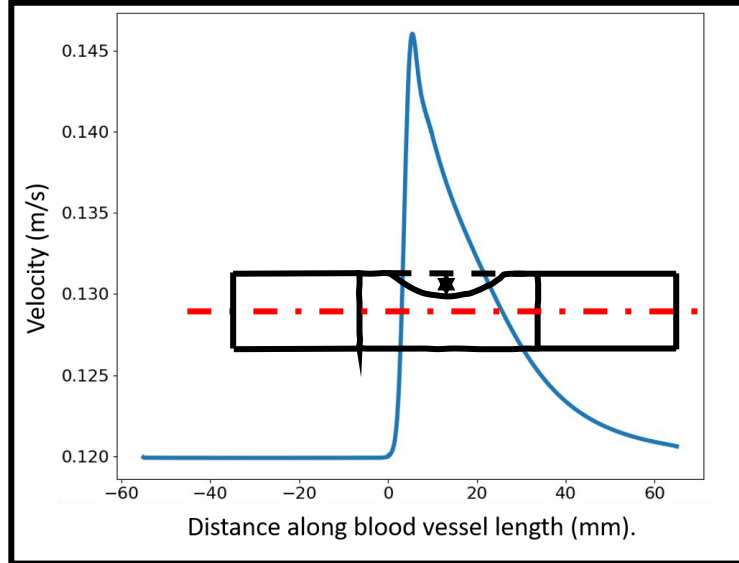
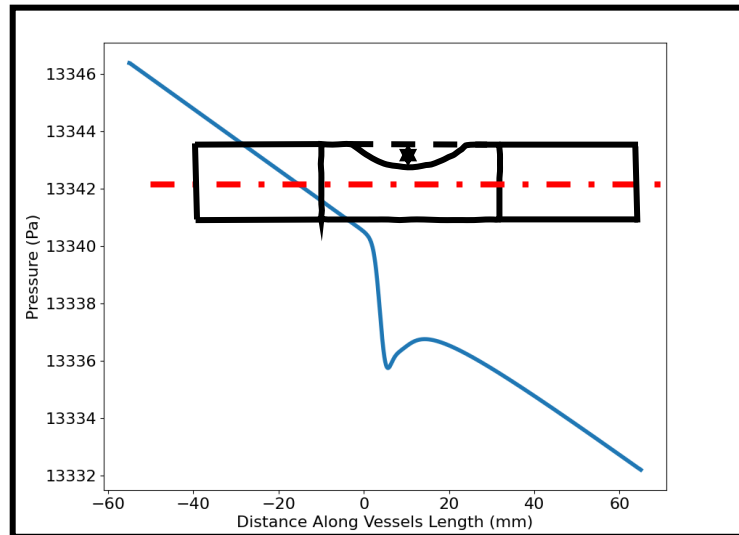


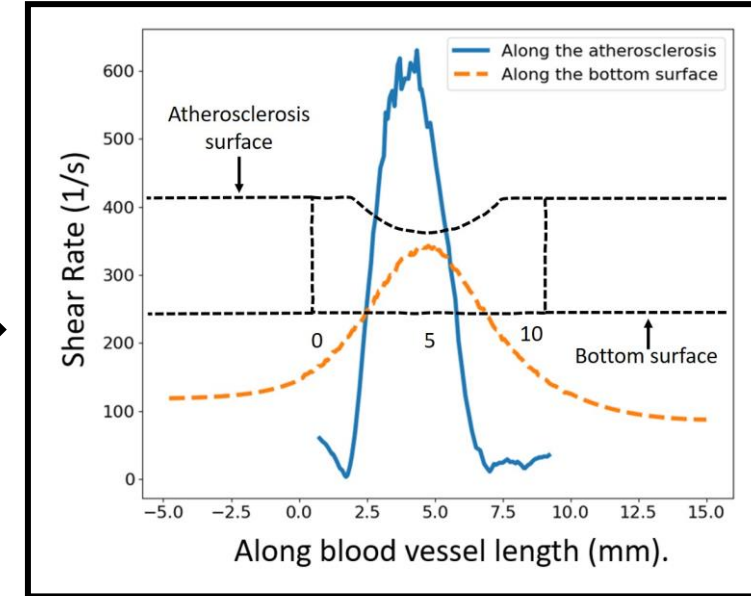
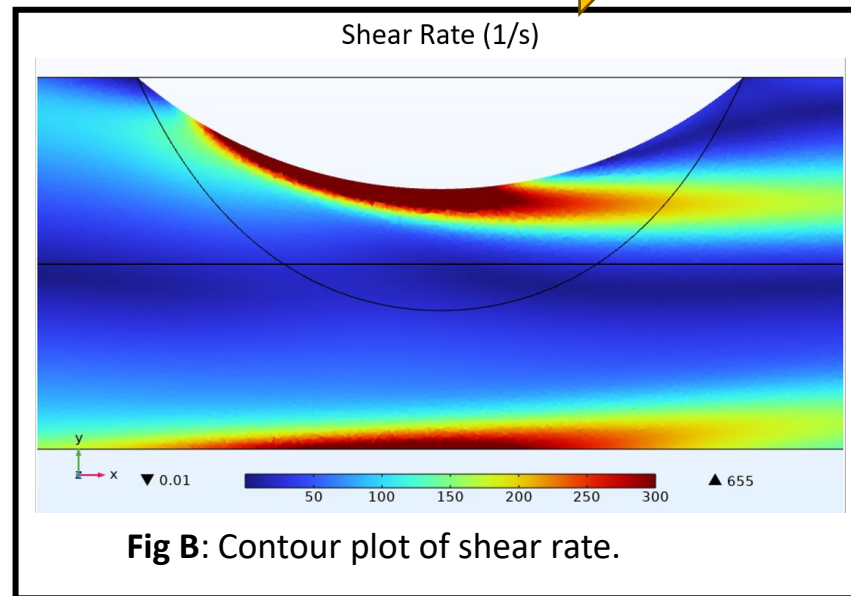
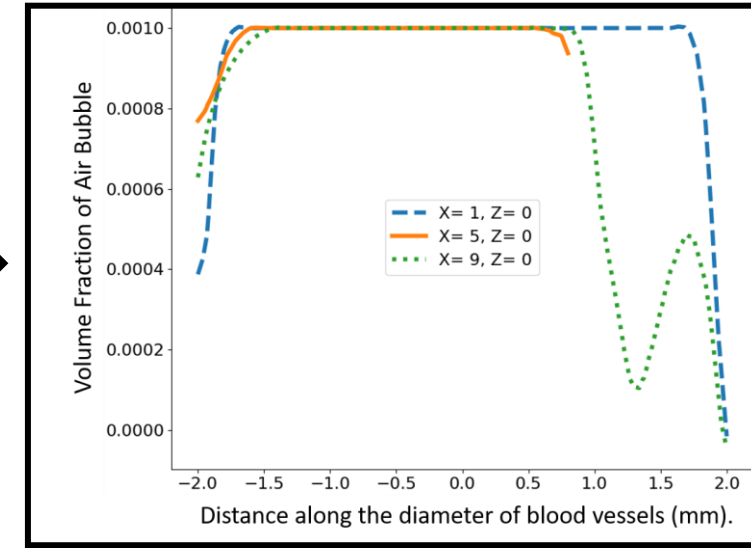
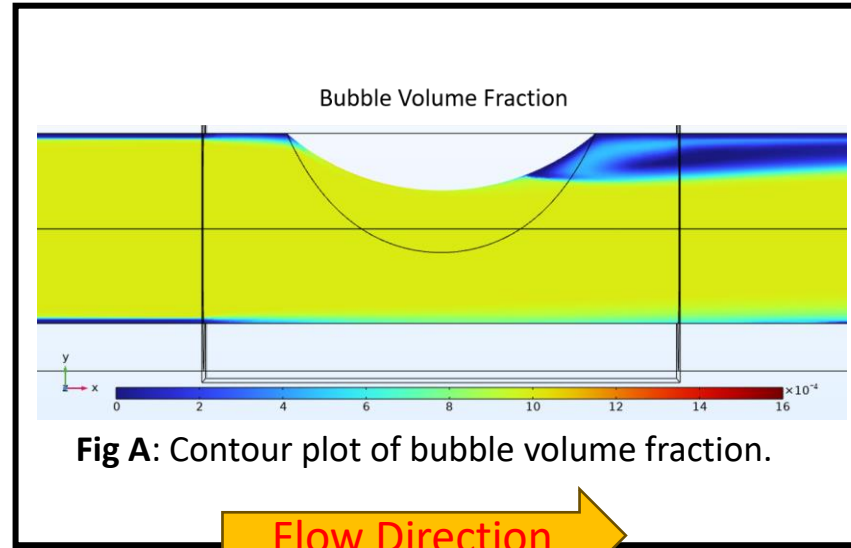
Fig B: Contour plot of pressure



- The velocity is increasing from 0.12 m/s to 0.15 m/s in the atherosclerosis region.
- The nature of flow deviated from the parabolic nature along the diameter of blood vessels.
- The pressure gradients are high in the atherosclerosis region.

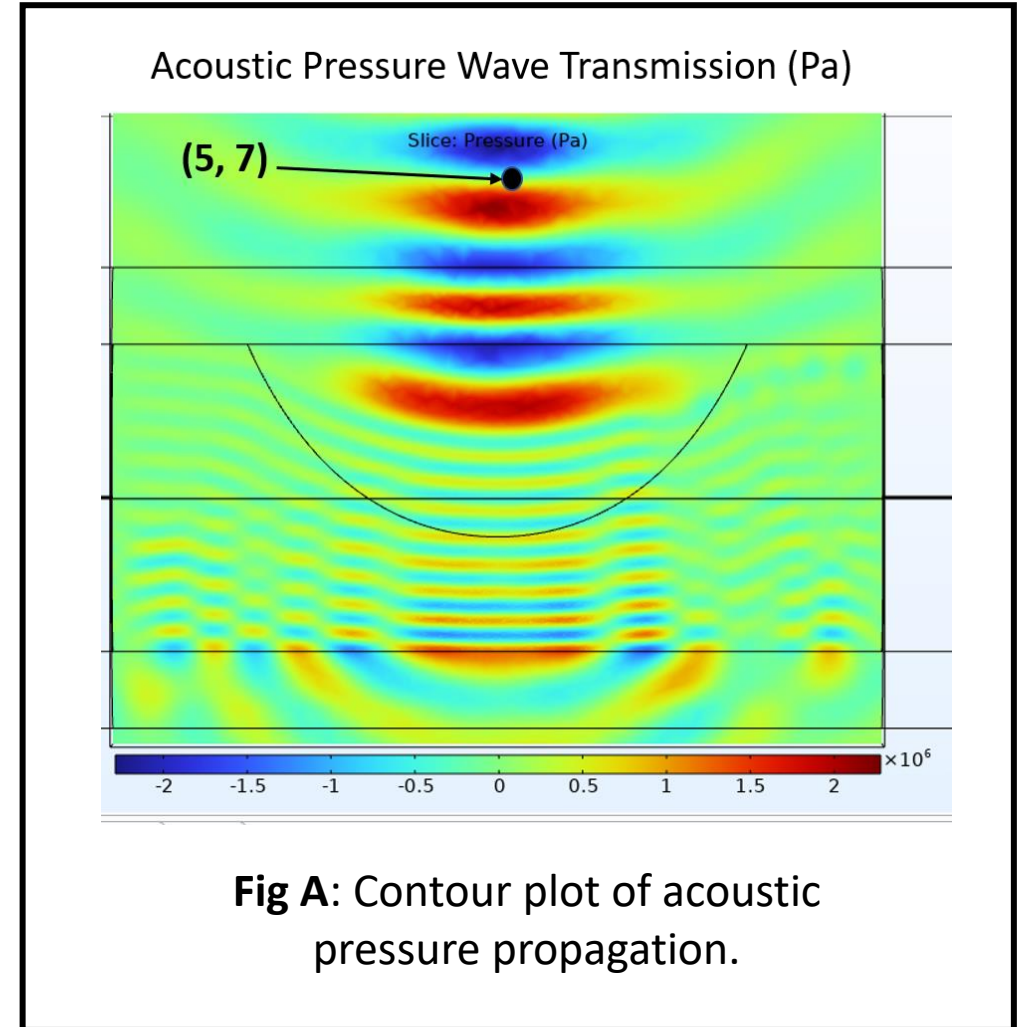
Numerical Results and Discussion-Flow

- Figure A shows the contour plot of the bubble distribution inside the blood vessels.
- The distribution of the bubble in the downward direction of the flow near the surface is 0.
- Figure B shows the contour plot of the shear rate inside the blood vessels.
- The solid line represents the shear rate along atherosclerosis, and the maximum is 655 s^{-1} .
- The dotted line represents the shear rate along the bottom surface, and the maximum is 335 s^{-1} .



Numerical Results and Discussion-Acoustic

- Figure shows the contour plot for the propagation of the acoustic wave.
- The mechanical index 1 ,and 1MHz frequency has been used.
- Acoustic signal has been recorded at the monitoring point (5,7,0) in the presence of the contrast bubble and in the absence of the contrast bubble inside the blood vessels.
- The sampling rate of the acoustic signal data is 10 times the frequency.
- Time step $\frac{1}{60f}$ has been used in the simulation.
- The root mean square (RMS) value of the signal was found to determine the effect of the contrast bubble on the strength of the echo.



Numerical Results and Discussion-Acoustic

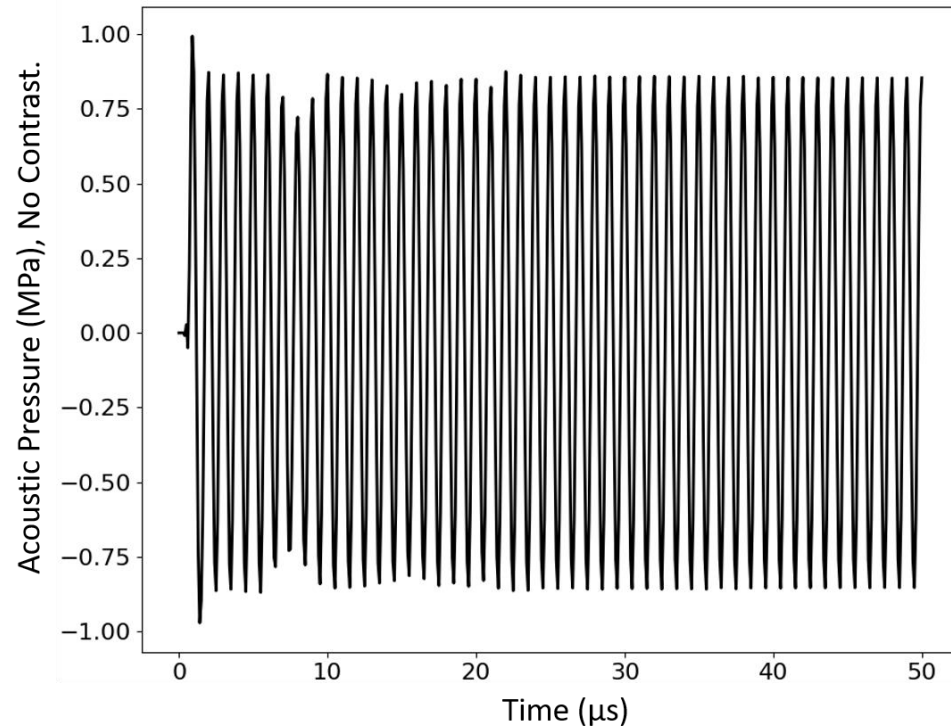


Fig A: Recorded acoustic signal at monitor point (5,7,0) in the absence of the contrast bubble.

- RMS value of acoustic signal in the absence of contrast bubble is 0.612 MPa.

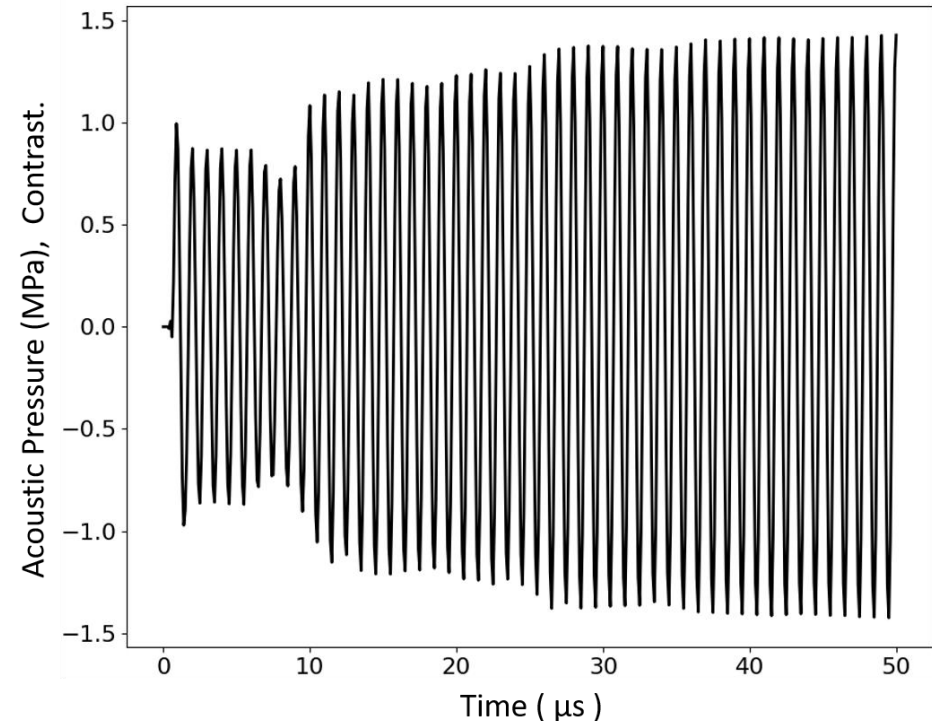


Fig B: Recorded acoustic signal at monitor point (5,7,0) in the presence of the contrast bubble.

- RMS value of acoustic signal in the presence of contrast bubble is 1.013 MPa.

- For 0.001 volume fraction of free air bubbles in water, acoustic impedance changing 1.5 MRayls to 0.35 MRayls. Also, for small volume fractions of air bubbles, there is a significant change in reflected acoustic pressure.

Experiment Setup and Sample Preparation

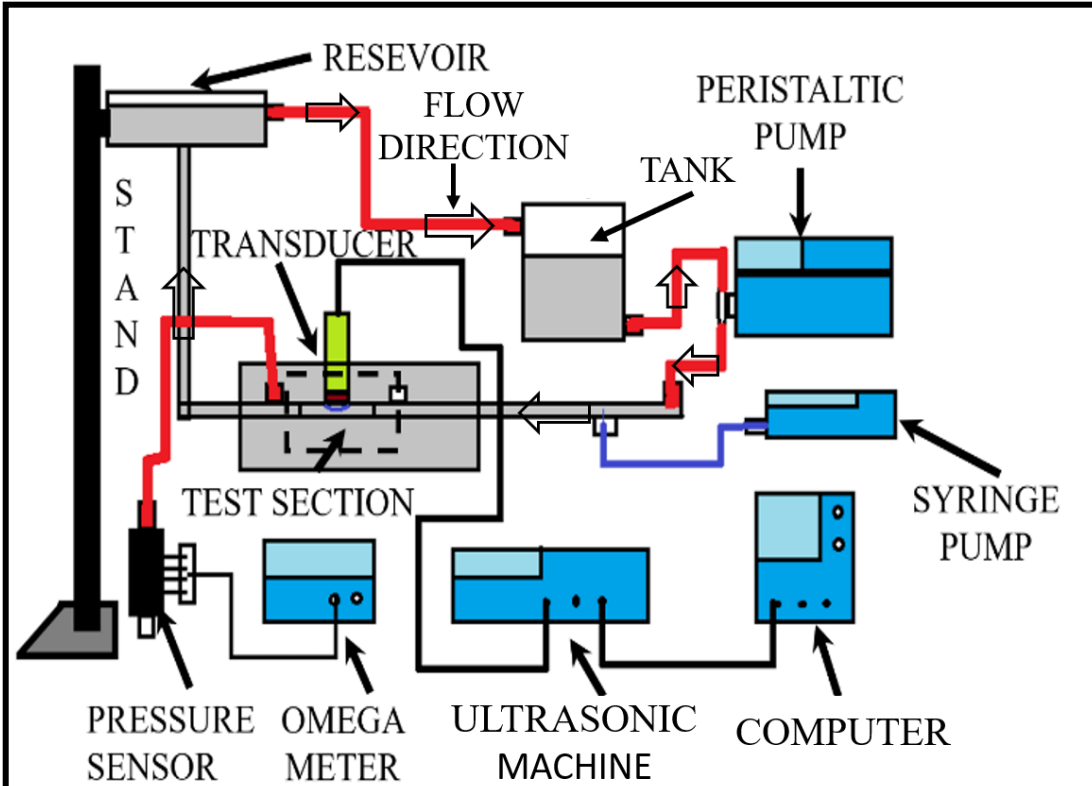


Figure A: Schematic diagram of experiments setup

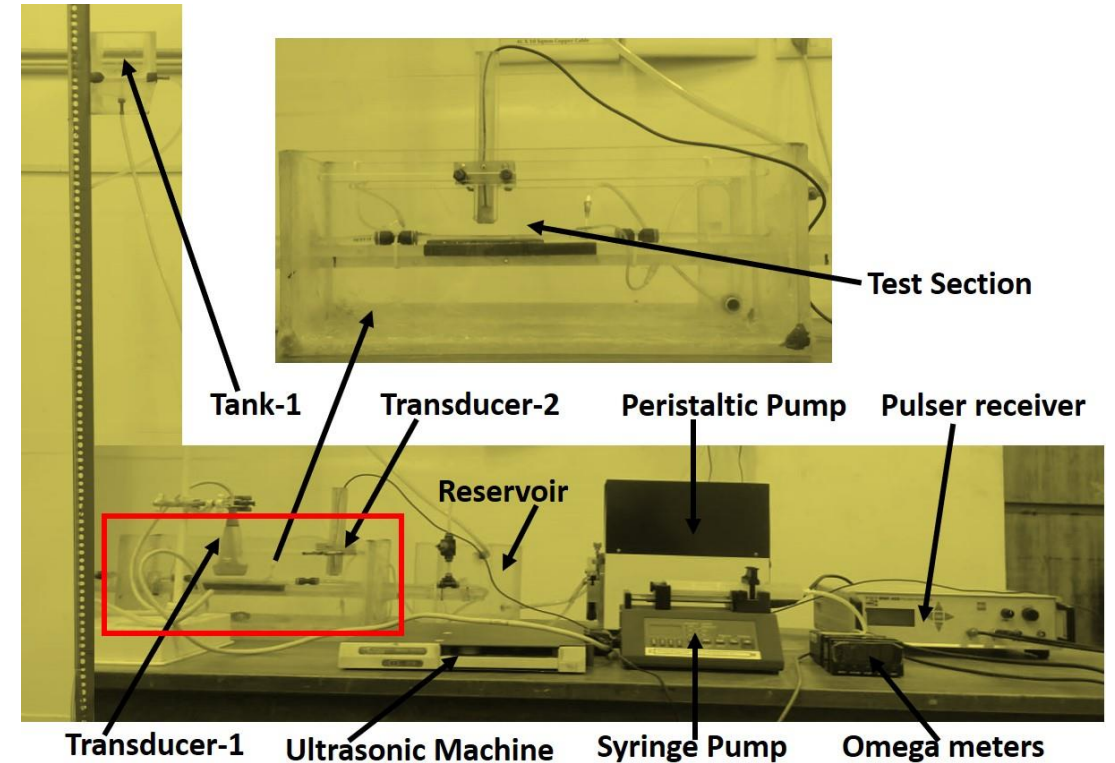
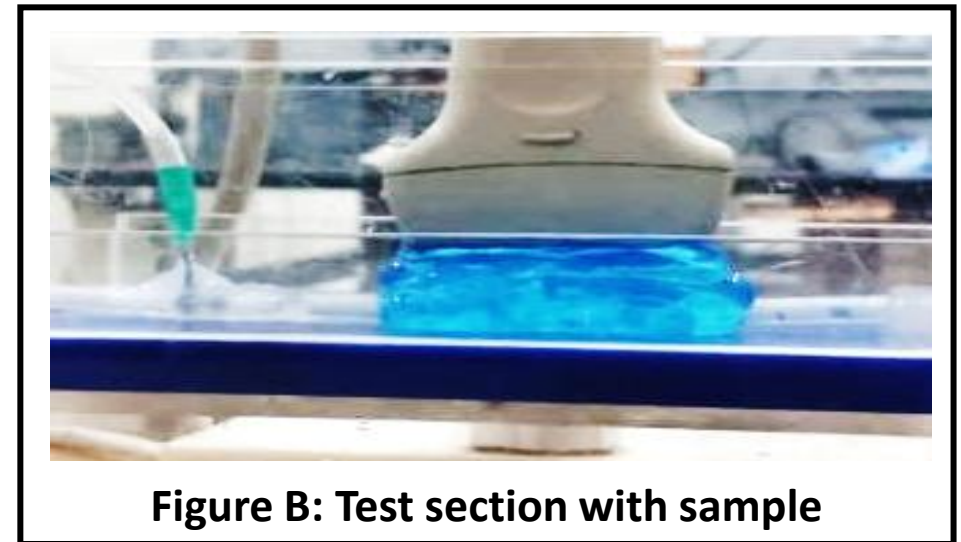
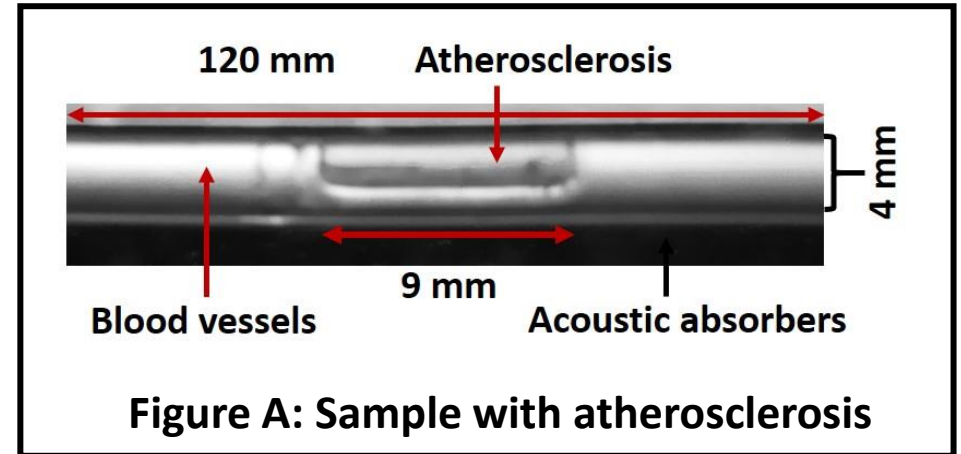


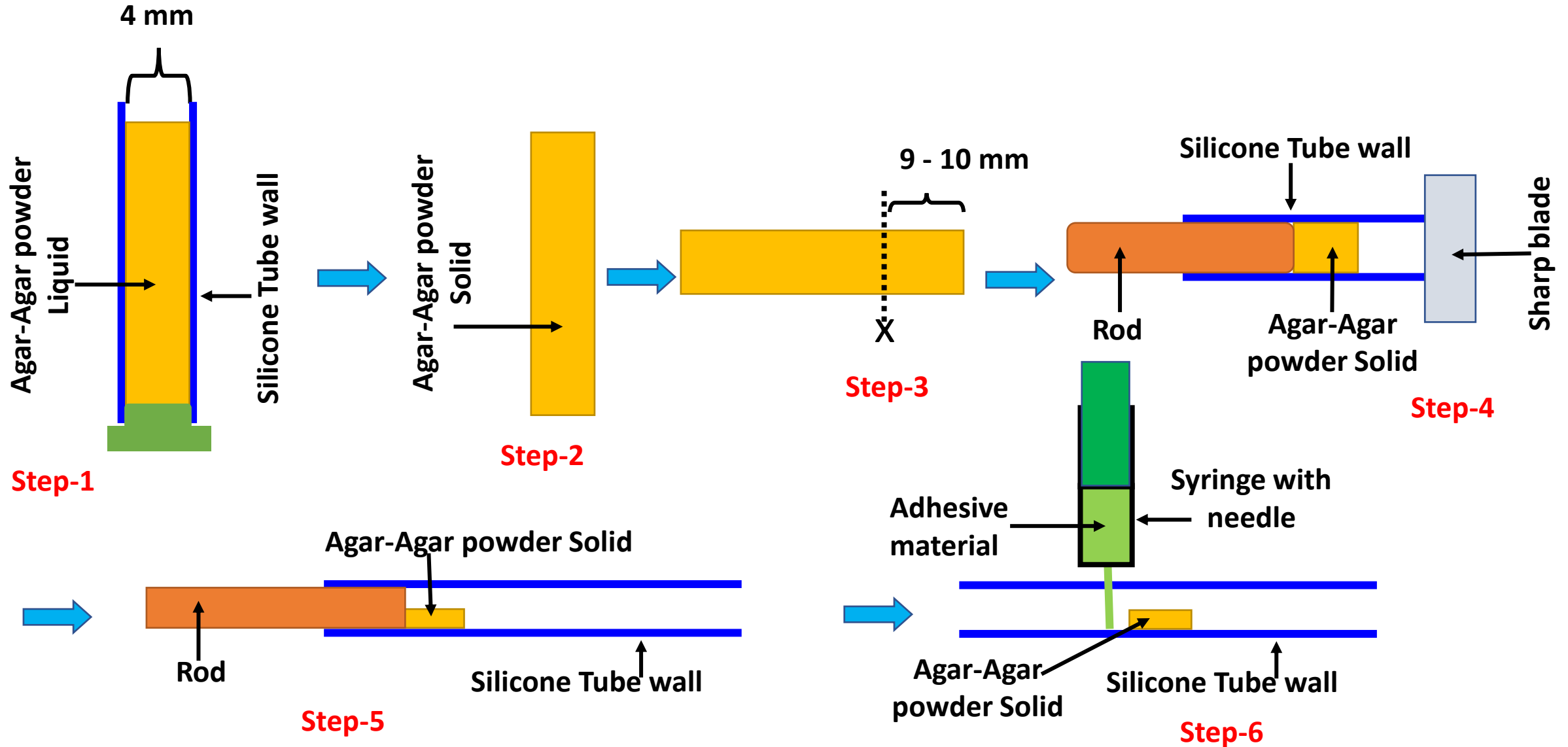
Figure B: Experiment setup

Experiment Setup and Sample Preparation

- Agar-agar powder has been used to make atherosclerosis.
- Acoustic impedance of agar-agar powder **1.56 MRayl**.
- The acoustic impedance of water is **1.50 MRayl**.
- The solution was prepared with **2 % Agar-agar** powder concentration in distilled water.
- Mixture solution is boiled with gentle stirring till the mixture becomes viscous in nature.
- Boiled solution is poured into a **4 mm** internal diameter tube (**as mold**) and kept for solidification.
- Solid agar-agar solution is taken out from the tube and cut into required dimensions to prepare atherosclerosis.
- Atherosclerosis shape agar-agar inserted into silicone rubber tube to prepare blockage.

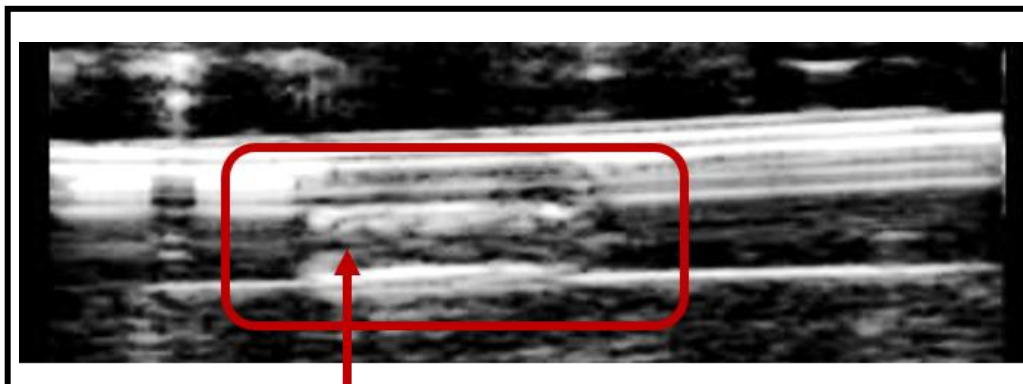


Experiment Setup and Sample Preparation



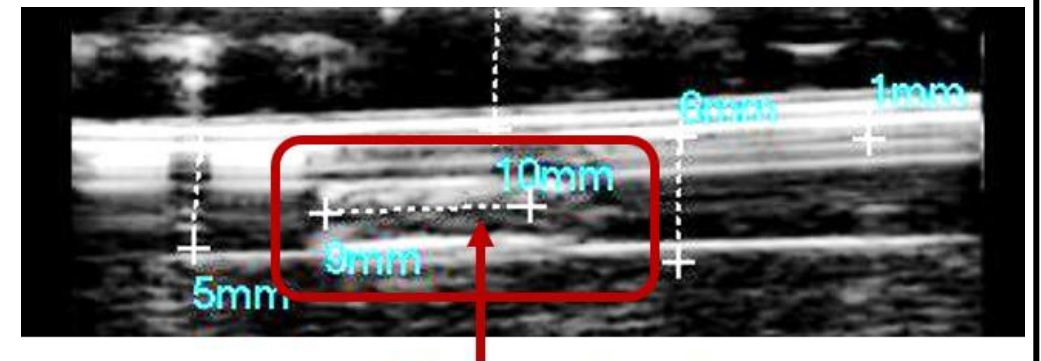
Experiments Results and Discussion-Air

- Dimension of atherosclerosis measured using ultrasonic imaging is consistent with the physical dimensions taken.
- Reflection of the acoustic wave is high at the interface of agar-agar powder and air due to a high mismatch of acoustic impedance at the interface.



Atherosclerosis zone

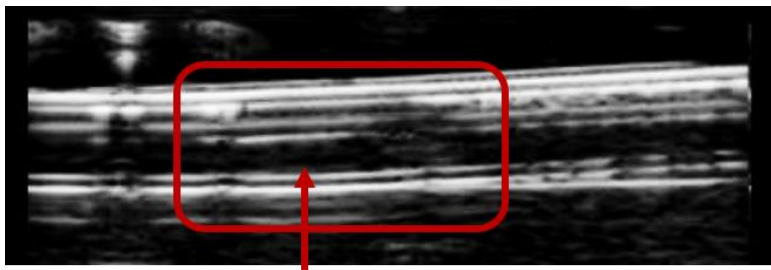
Fig A: Ultrasonic image of atherosclerosis blood vessels, where blood vessels are filled with air.



Atherosclerosis zone

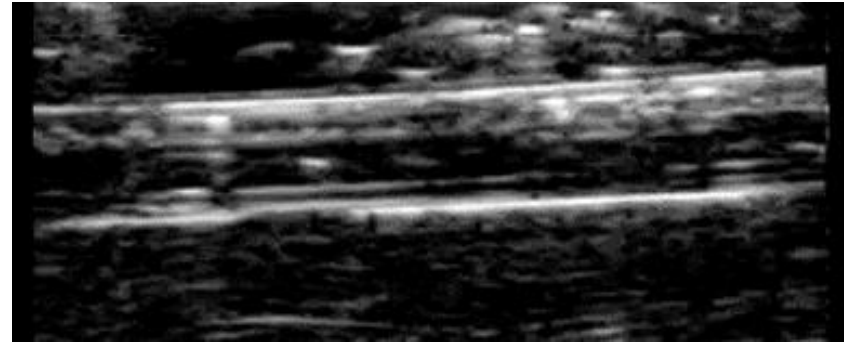
Fig B: Ultrasonic image to characterize the dimension of atherosclerosis, where blood vessels are filled with air.

Experiments Results and Discussion-Saline Water

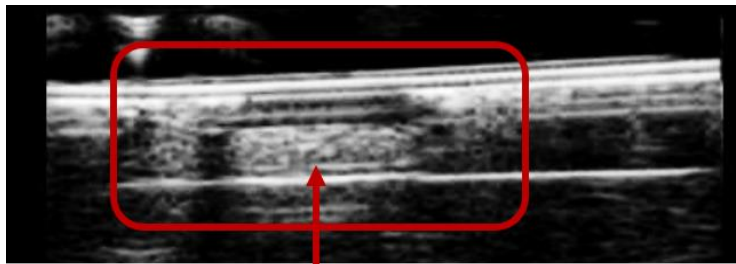
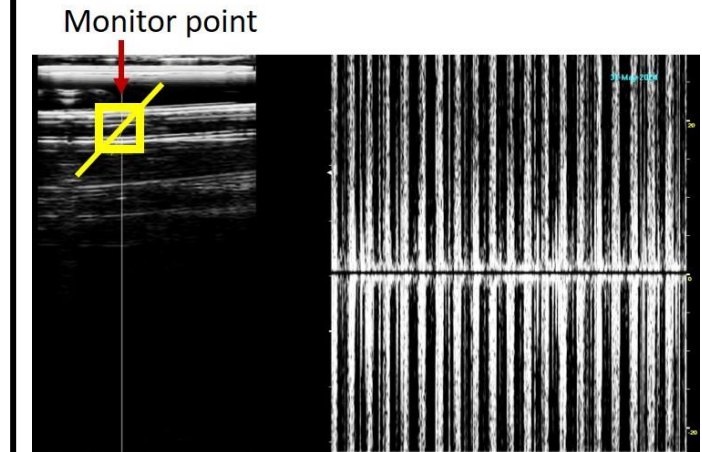


Atherosclerosis zone

Fig A: Ultrasonic image, where flowing fluid is saline water, No bubble inserted. Flow is right to left.

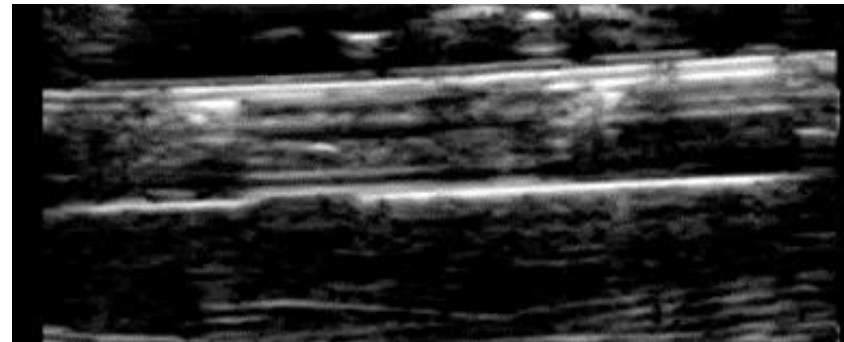


Flow Direction

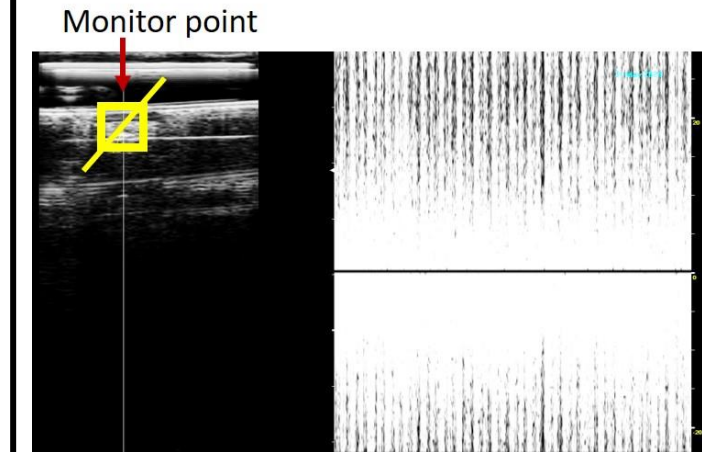


Atherosclerosis zone

Fig B: Ultrasonic image, where flowing fluid is saline water, bubble inserted. Flow is right to left.



Flow Direction



Conclusions

- Velocity increases from 0.12 m/s to 0.15 m/s in the atherosclerosis region.
- Sharp changes in pressure occur near the blockage area, due to which more shear force is generated at the blockage region.
- The maximum displacement of the inner wall of blood vessels is 35 μm , and it varies linearly along the radius of blood vessels.
- The maximum shear rate on the atherosclerosis surface is 655 [1/s], and on the bottom surface, 335 [1/s].
- For small volume fraction of contrast bubbles, there is significant change in strength of echo produce at the interface of atherosclerosis and blood.
- The RMS value of the acoustic signal in the absence of the contrast bubble is 0.612 MPa, and in the presence of the contrast bubble, it is 1.013 MPa.
- In-vitro experiment results and numerical simulation results show that the method can be used to improve US imaging for the diagnosis of early atherosclerosis.

Future Work

- Developed Ultrasound contrast microbubbles of diameter 1 – 5 μm .
- Repeat the same experiments using in-house developed microbubbles.
- Find quantitative results in experiments to find the effects of the microbubbles on the strength of the echo.

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Publications

Conference Paper: **Presented**

[1]. A. Kumar, D. Chatterjee, and S. Vengadesan, 2022-23. Towards an Early Detection of Atherosclerosis in Human Coronary Artery using Ultrasound Contrast Agent: Numerical Simulation. [Conference on cavitation Exploitation, Ljubljana 2022-2023, Slovenia].

Journal Paper: **Submitted** [Manuscript Number: ULTRAS-D-24-00511, **Ultrasonics**]

[1]. A. Kumar, D. Chatterjee, and S. Vengadesan, 2024. Towards an Early Detection of Atherosclerosis in Human Coronary Artery using Ultrasound Contrast Agent: Numerical Simulation and Experiments work.

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Thank you

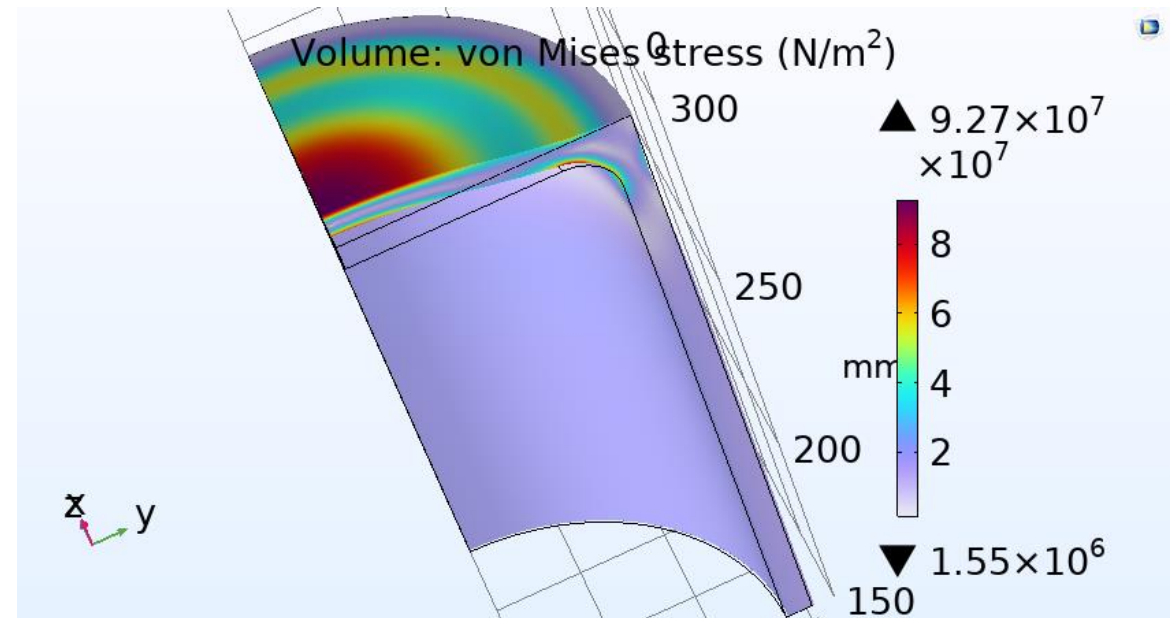
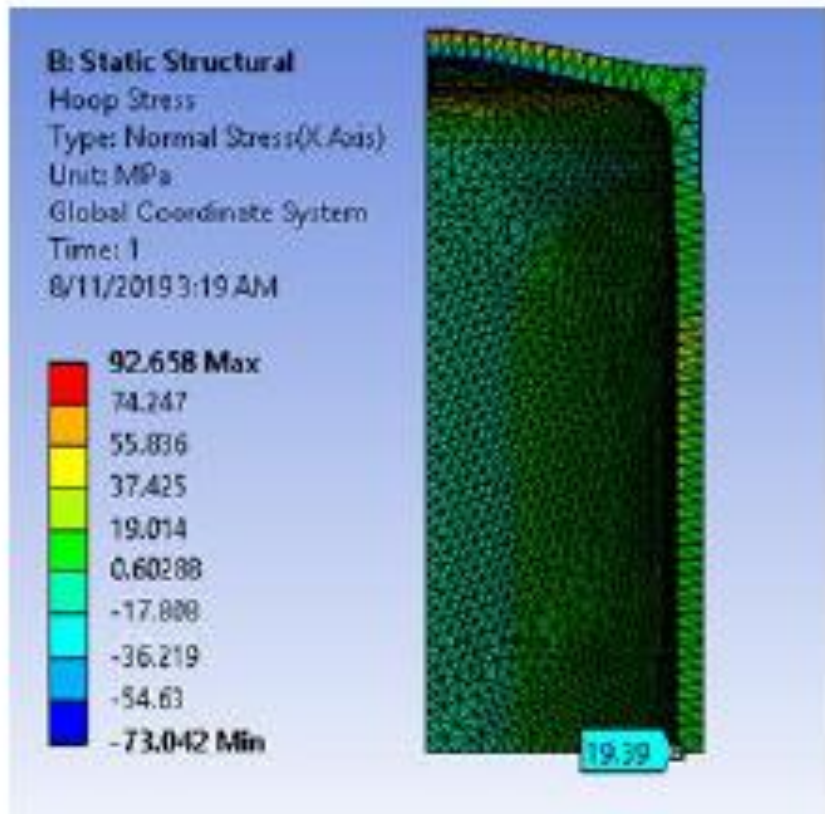
Additional

$$\mathbf{K} = \mu_{\text{app}}(\nabla \mathbf{u} + (\nabla \mathbf{u})^T)$$

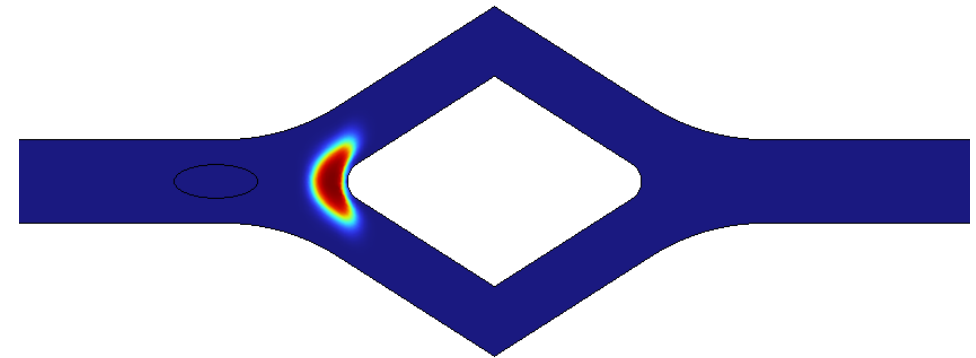
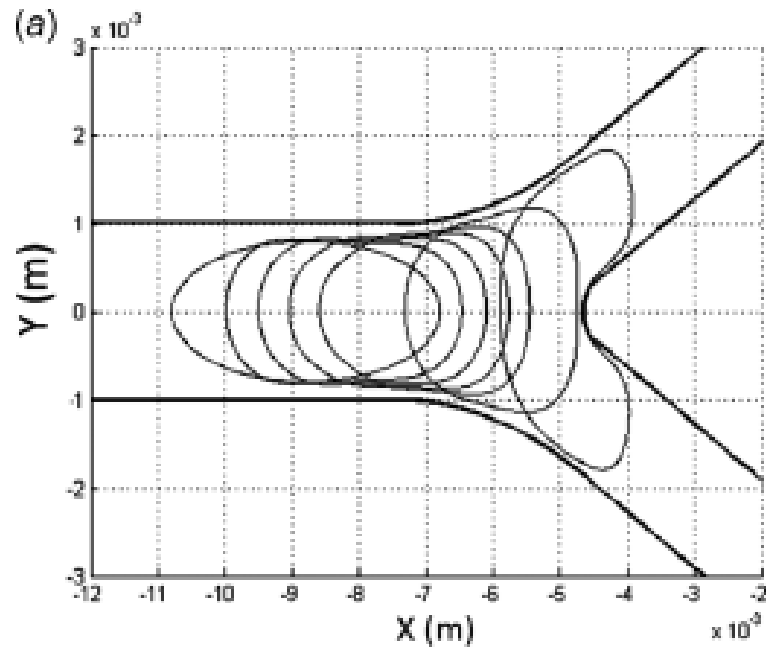
$$\mu_{\text{app}} = \left[\sqrt{\mu_p} + \sqrt{\frac{\tau_y}{\dot{\gamma}}} \left[1 - \exp\left(-\sqrt{m_p \dot{\gamma}}\right) \right] \right]^2$$

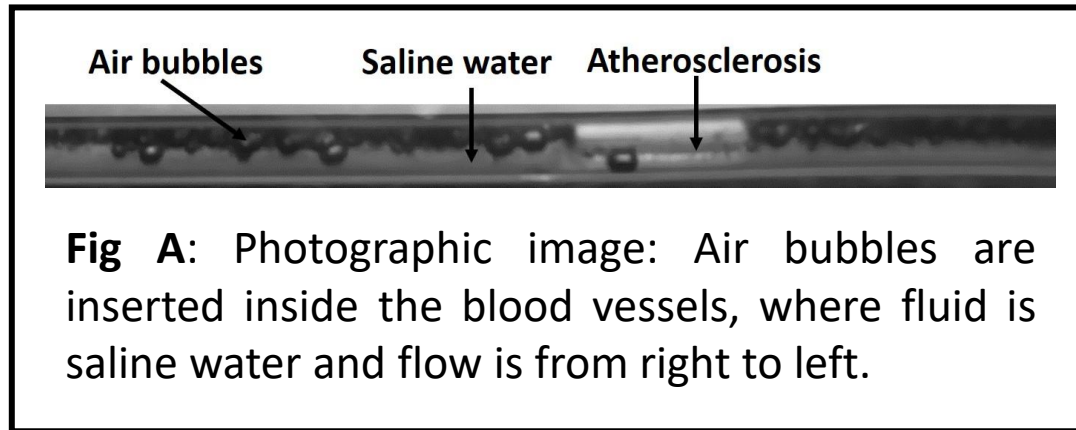
$$\dot{\gamma} = \sqrt{2\mathbf{S} : \mathbf{S}}, \quad \mathbf{S} = \frac{1}{2}[\nabla \mathbf{u} + (\nabla \mathbf{u})^T]$$

stress

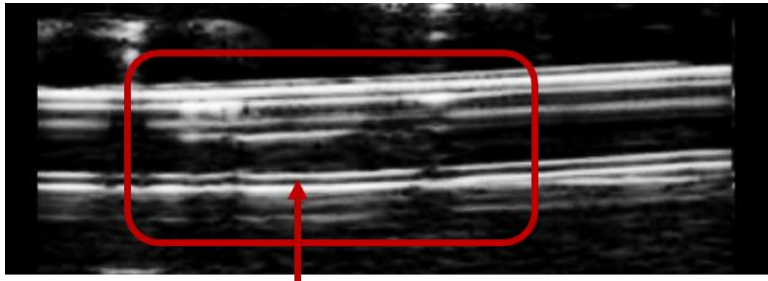


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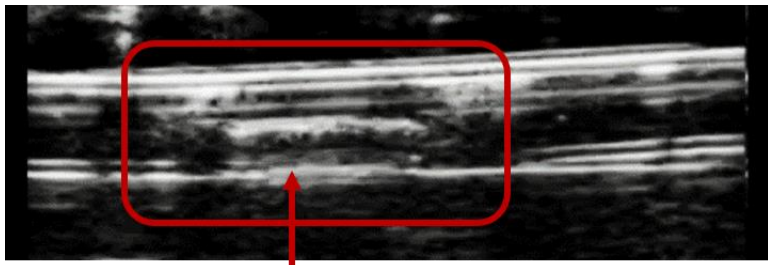
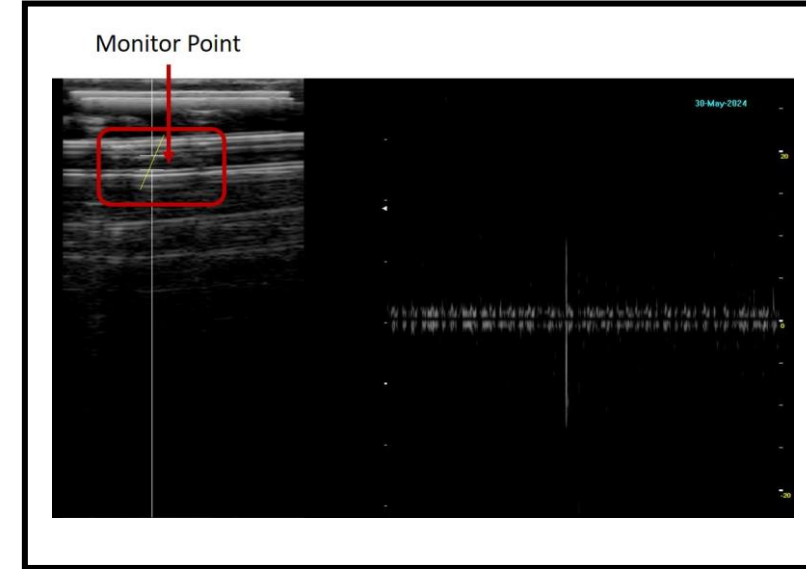
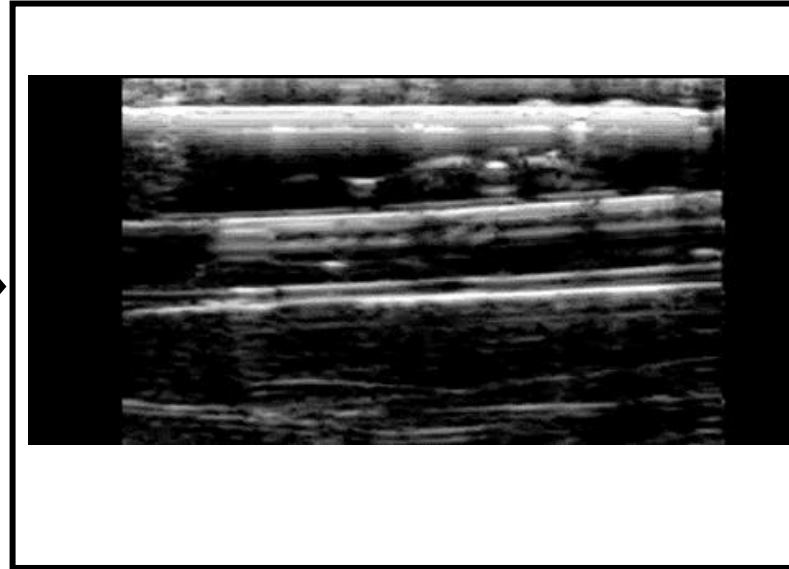


Experiments Results and Discussion-Water



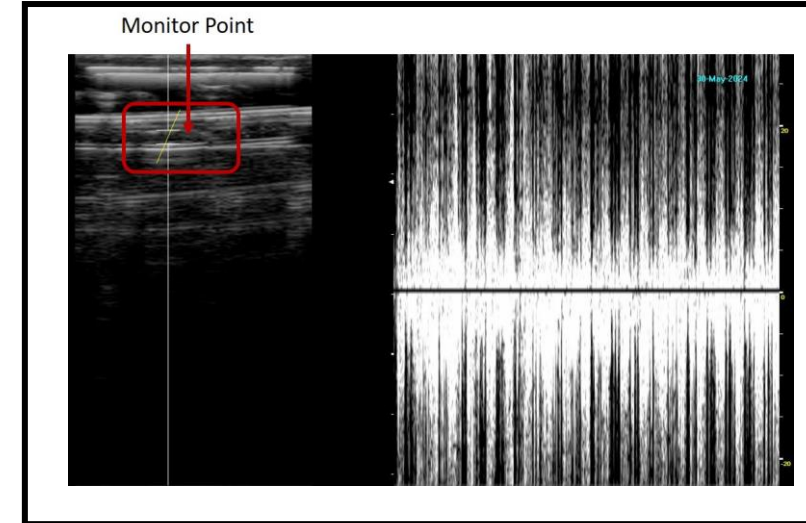
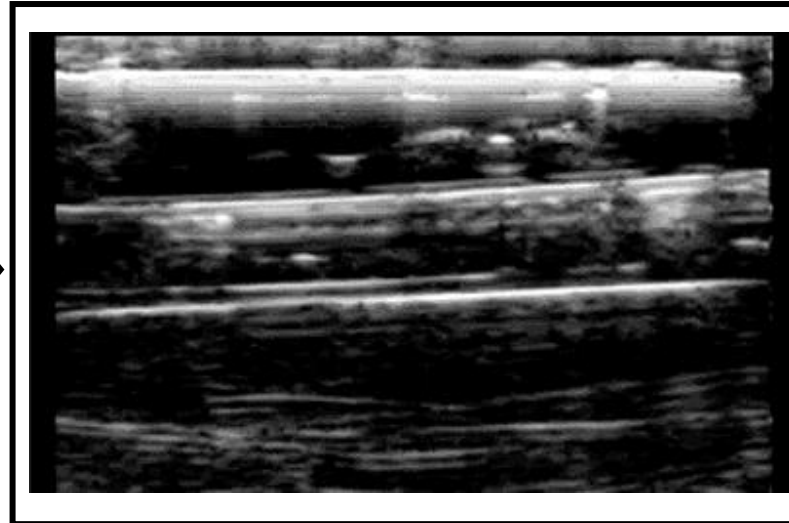
Atherosclerosis zone

Fig A: Ultrasonic image, where flowing fluid is water, No bubble inserted. Flow is right to left.

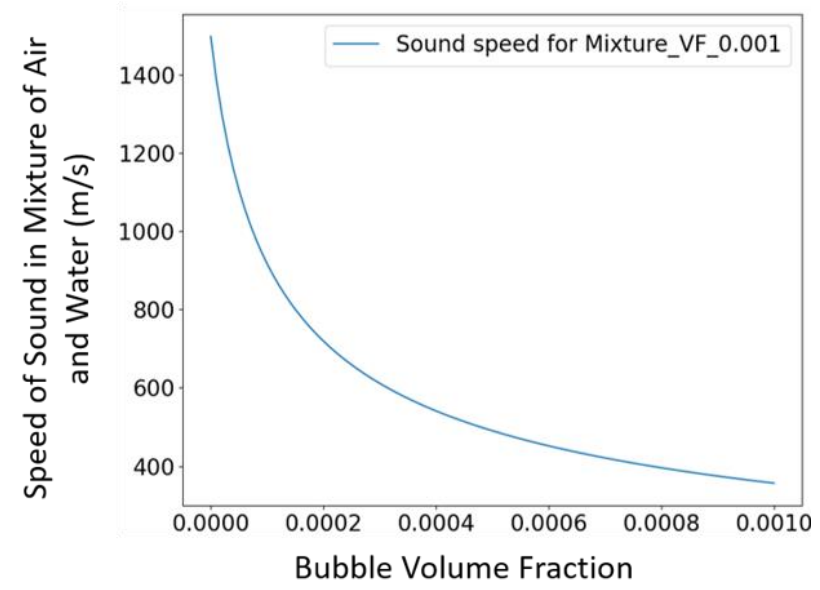
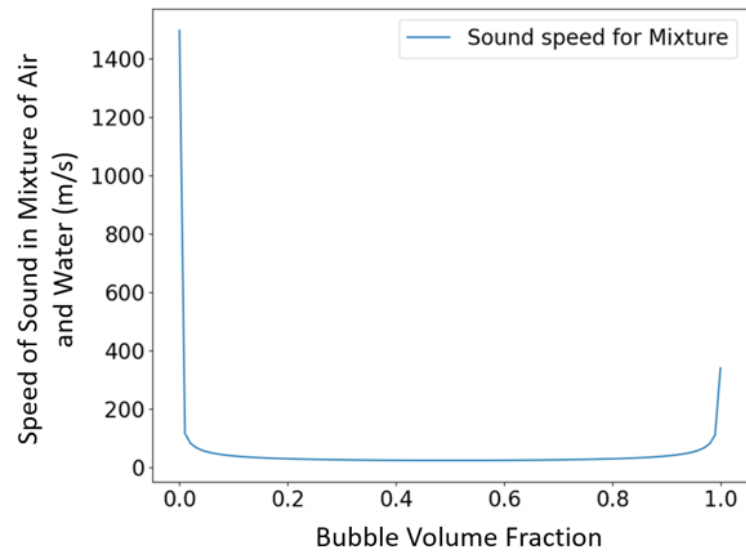


Atherosclerosis zone

Fig B: Ultrasonic image, where flowing fluid is water, bubble inserted. Flow is right to left.



- Methods available for the diagnosis of atherosclerosis, such as evaluation of electrocardiogram (ECG), invasive angiography, magnetic resonance coronary angiography (MRCA), ultrasound, computer tomography (CT), and single-photon emission computed tomography (SPECT), and Intravascular ultrasound imaging (IVUS).
- The main component is lipid accumulation, and it may reduce the blood flow rate by up to 50% (Fayad and Fuster, 2001).



$$1 \text{ Pa} = 1 \text{ kg} \cdot \text{m}^{-1} \cdot \text{s}^{-2}.$$

As an MKS unit, one rayl equals one pascal-second per meter ($\text{Pa} \cdot \text{s} \cdot \text{m}^{-1}$), or equivalently one newton-second per cubic meter ($\text{N} \cdot \text{s} \cdot \text{m}^{-3}$).

Formula

$$c = \sqrt{\frac{K_s}{\rho}}$$

c = speed of sound

K_s = coefficient of stiffness

ρ = density

$$\bar{\tau} = \frac{64}{49}\tau_c + \frac{4Qk}{\pi R^3} + \frac{32}{7}\sqrt{\frac{kQ\tau_c}{\pi R^3}}$$

Solid mechanics equation.

$$\nabla \cdot (FS)^T + F_v = 0 \quad (5)$$

$F = I + \nabla u_{solid}$, S= Stress, F= Deformation gradients, F_v = Volume force, u= Solid wall velocity.

Here, the thickness of blood vessels is 1 mm, and the diameter is 4 mm; blood vessels have been considered as thick ones with $t/d = 25\%$. Because of this, the stress will vary in the radial direction. To capture this, Lamé's parameters have been used to find stress in the radial direction.

Lamé's first parameter λ_1

$$\lambda_1 = \frac{E\nu}{(1-\nu)(1-2\nu)} \quad (6)$$

Lamé's second parameter λ_2

$$\lambda_2 = \frac{0.5E}{(1+\nu)} \quad (7)$$